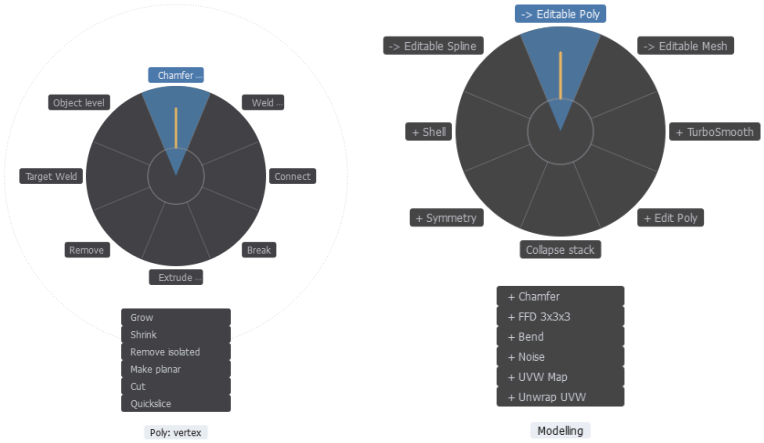
# MarkingForge

Marking menus, hotbox and gesture sliders for Autodesk 3ds Max 2027

INSTALLATION AND CONFIGURATION GUIDE · VERSION 1.5.108




## REAL MARKING MENUS - A QUICK GESTURE NEEDS NO DIAL

MarkingForge works the way marking menus are meant to: the command is chosen by the **direction** of the movement, not by clicking an item. Press the key and flick the mouse towards the command in one quick movement - the command runs and **the dial is not even drawn**. The dial appears when you hold the key with the mouse still - after a tenth of a second - to remind you where things are. After a few days your hand knows the directions and you stop looking.

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Hold a key, flick the mouse, done. This guide covers installing the plugin from scratch, the first start, setting up shortcuts, using the 24 built-in dials, changing them, adding MarkingForge commands to 3ds Max's menus, toolbars and quad menus, and updating and uninstalling. The menu editor part by part is in the **MarkingForge Editor Guide**; every option in detail, with tips and troubleshooting, in the **MarkingForge Reference**.

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# New since 1.5.70 - read this first

What changed how MarkingForge feels - each takes a second to try. Questions and ideas: the MarkingForge Discord, **https://discord.gg/HsakZWQZAw**.

### You never have to hit a tile to confirm

A dial picks by the **direction** you move in, not by the caption under the cursor. Move towards a command and let go anywhere that way - near the centre, on its caption or far past it: the command runs. Only the **centre** cancels. In Maya's shape the caption under the cursor wins.

### The dial opens in a tenth of a second (1.5.72)

Hold the key with the mouse still: the dial is there after 100 ms - 3ds Max alone waits 250 ms. **Dial speed...** in the editor sets 0 to 250 ms and an opening animation. A flick - moving at once - never waits: the command runs and no dial is drawn.

### Your size (1.5.70, 1.5.73)

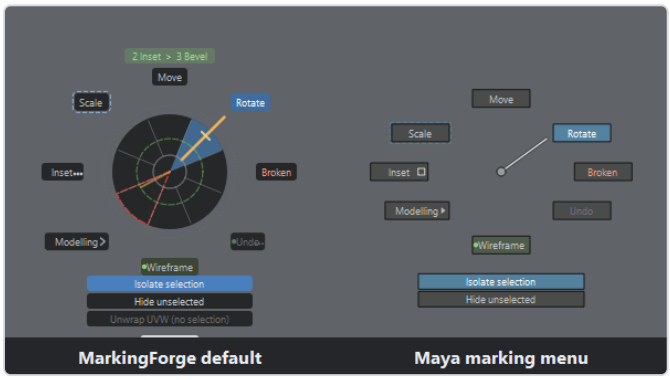
**Dial size...** in the editor: the whole dial 60 to 160 %, the text 70 to 200 %, and the green return ring of multiple picks - with a picture of the real dial while you drag.

### Any key - Tab too (1.5.73, 1.5.75)

A dial can sit on any key. On Tab it opens from the viewport and the panels alike and never walks the keyboard into a number field. Ctrl+Z silent while the toolbar's Undo works? The keyboard is in a field: press Esc.

### Exactly Maya's marking menu (1.5.72 - 1.5.74)

One tick: **MarkingForge > Experimental features > Maya marking menu look**. Every dial is then drawn as in Maya: no disc, grey boxes round a small centre dot, a line to the cursor, Maya's blue on the box you aim at, and Maya's **option boxes** - let go on the small square beside a command to open its settings. Clear the tick and your own colours are back. For one dial only: **Colours...** > preset *Maya marking menu*, or just its **Shape** list.




The same dial in both shapes: the MarkingForge dial (left) and the Maya marking menu look (right) - Maya's option box beside *Inset*.

### Newest (1.5.80 - 1.5.101)

**Chain submenus** (1.5.98, always on since 1.5.102): a submenu opens under the cursor and leaves a circle to slide back to. **Instant over heavy objects** (1.5.101): the menu under the cursor no longer waits for 3ds Max on a mesh of millions of faces. **A tap repeats your last modelling command** (Experimental features > Tap repeats the last modelling command). **Right-click cancels** a held dial; a quick tap runs nothing.

# 1. What you get

The download is one ZIP file. Unpacked, it holds:

```
MarkingForge_1.5.108\
  MarkingForge_1.5.108.mzp                  installer - drag onto a viewport (method 1)
  Documentation\
    MarkingForge_Installation_Guide.pdf     this document
    MarkingForge_QuickStart_Long.pdf        22 lessons: first gestures, the editor
    MarkingForge_QuickStart_Short_v3.pdf    the short version - 16 lessons and the tips, about 15 minutes
    MarkingForge_Editor_Guide.pdf           the menu editor, part by part
    MarkingForge_Reference.pdf              every option, tips, troubleshooting
    MarkingForge_Shortcut_Card.pdf          the 24 dials on one page to print
    MarkingForge_for_Maya_Users.pdf         Maya habits mapped to MarkingForge
    MarkingForge_Preset_Packs.pdf           the four ready-made dial sets
    MarkingForge_Studio_Deployment.pdf      silent install for studios (method 4)
  Dials\Built-in, Extra, Packs\             every dial as a .mfdial file (24 + 12 + 15)
  Presets\MarkingForge_Pack_*.json          Modelling, Animation, Look-dev, Archviz
  Studio\Deploy-MarkingForge.ps1            silent install / update / removal (method 4)
  README.txt                                the short version of this document
  CHANGELOG.txt                             what is in this version
  LICENSE.txt                               licence agreement
  Manual install\
    MarkingForge_Install.ms                 the same installer as a script (method 2)
    MarkingForge\                           the plugin folder itself (method 3)
      PackageContents.xml                   tells 3ds Max what to load
      plugins\MarkingForge.gup              the plugin (C++)
      scripts\MarkingForge_Startup.ms       builds the MarkingForge menu
      macroscripts\MarkingForge_Macros.mcr  the menu commands
      tools\mf_editor\*.py                  the menu editor (Python)
      dials\                                the dial library the editor reads
      Documentation\*.pdf                   these guides - MarkingForge > Documentation
```

The folder `Manual install\MarkingForge` is exactly what every installation method puts on your disk; the .mzp carries the same folder inside it.

### What kind of plugin is it?

MarkingForge is a **native C++ plugin** - a Global Utility Plugin, the `.gup` file, which is a Windows DLL that 3ds Max loads when it starts. The dials, the gesture recognition and the drawing all run in C++ with Qt, the same toolkit 3ds Max itself is built with, so a dial appears in a few milliseconds.

Around it sit three small parts: a **startup script** (MAXScript) that adds the MarkingForge menu to the menu bar, a set of **macroscripts** (MAXScript) behind the menu commands - which is also what lets you put them on toolbars and quad menus - and the **menu editor**, written in Python with PySide6, both of which ship with 3ds Max 2027. Next to them lies the **dial library**: every built-in, extra and pack dial as a file of its own, which the editor reads. Nothing else needs installing.

> Everything is installed as one folder that 3ds Max loads through its standard *Application Plugins* mechanism. Nothing is copied into the 3ds Max program folder, and nothing is installed outside your user profile unless you choose "All users".

# 2. Requirements

| Item | Requirement |
|---|---|
| 3ds Max | Autodesk 3ds Max 2027, Windows 64-bit. This download is for 3ds Max 2027 only - it does not load in other versions, and the installer checks and refuses. MarkingForge is available for 3ds Max 2025-2027, each version with its own download; 3ds Max 2024 and earlier may follow if enough users ask for them. |
| Windows | Windows 10 or 11, 64-bit. |
| Visual C++ runtime | Microsoft Visual C++ 2015-2022 Redistributable (x64). 3ds Max installs it itself, so it is almost always present. If the plugin does not load, reinstall it from Microsoft ( `vc_redist.x64.exe` ). |
| Rights | None special for "Current user". "All users" needs 3ds Max started as administrator (or, for methods 3 and 4, an administrator's permission in Windows). |
| Studio script (method 4 only) | Windows PowerShell, which is part of Windows 10 and 11. |

# 3. Installing from scratch

Installing MarkingForge means putting one folder, `MarkingForge`, into Autodesk's *Application Plugins* folder, where 3ds Max looks for plugins every time it starts. There are four ways to do it and all of them end in the same place - pick the one that suits you. Methods 1 and 2 run the same installer; method 3 is the same copy done by hand; method 4 is the same copy for studios, without a window.

| Installed for | Folder | Rights |
|---|---|---|
| You only (recommended) | `%APPDATA%\Autodesk\ApplicationPlugins\MarkingForge` | none special |
| Every user of the computer | `%ProgramData%\Autodesk\ApplicationPlugins\MarkingForge` | administrator |

Install it in ONE of the two, never both: 3ds Max would find the plugin twice and report a duplicate. The installer and the studio script both check for that.

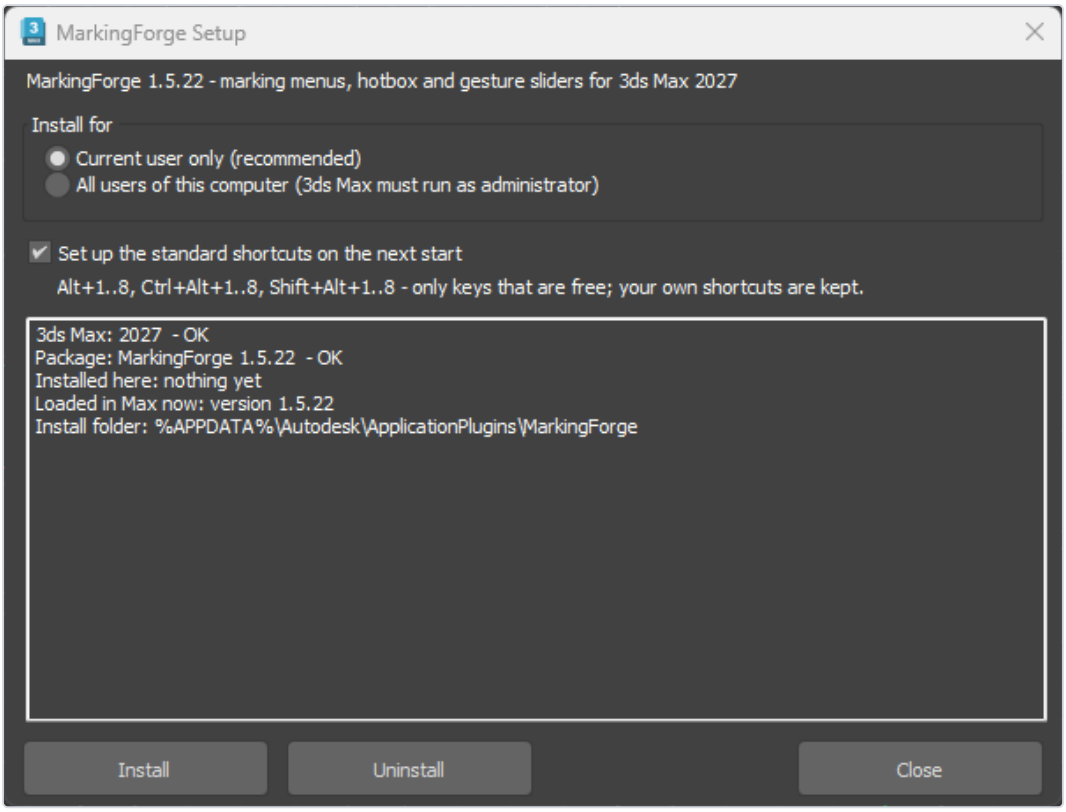
### Before you start

1. Unzip the whole download into a folder of your own. Every method works from the unzipped files - the installer copies the `MarkingForge` folder that lies next to it.
2. Check the version: this download is for **3ds Max 2027**. Every other 3ds Max version has a download of its own; the installer refuses a 3ds Max it was not built for.
3. If a copy of `MarkingForge.gup` was ever put into 3ds Max's own `plugins` folder by hand, remove it. The Setup window lists every copy it finds.

| Method | Use it when | 3ds Max |
|---|---|---|
| 1 - drag the .mzp onto a viewport | You install on your own computer - the usual case. | running |
| 2 - run the installer script | The same installer, started from the Scripting menu. | running |
| 3 - copy the folder by hand | You want no installer at all. | closed |
| 4 - the studio script | Many computers, deployment tools, log-on scripts - no dialog. | closed |

### Method 1 - drag and drop the .mzp RECOMMENDED

1. Unzip the download and start 3ds Max 2027.
2. Drag `MarkingForge_1.5.108.mzp` from Explorer and drop it onto any viewport. (Choosing the .mzp in **Scripting > Run Script...** does the same.)
3. The **MarkingForge Setup** window opens. Check the lines in its status box - each should end in *OK*. Leave **Current user only (recommended)** selected and press **Install**.
4. Restart 3ds Max.



The MarkingForge Setup window. It checks the 3ds Max version, the package and any other copy of the plugin before it lets you install.

### Method 2 - run the installer script

1. Unzip the download.
2. In 3ds Max choose **Scripting > Run Script...**
3. Open `Manual install\MarkingForge_Install.ms`. Leave it where it is: it installs the `MarkingForge` folder beside it.
4. The same Setup window opens - press **Install**, then restart 3ds Max.

### Method 3 - copy the folder yourself

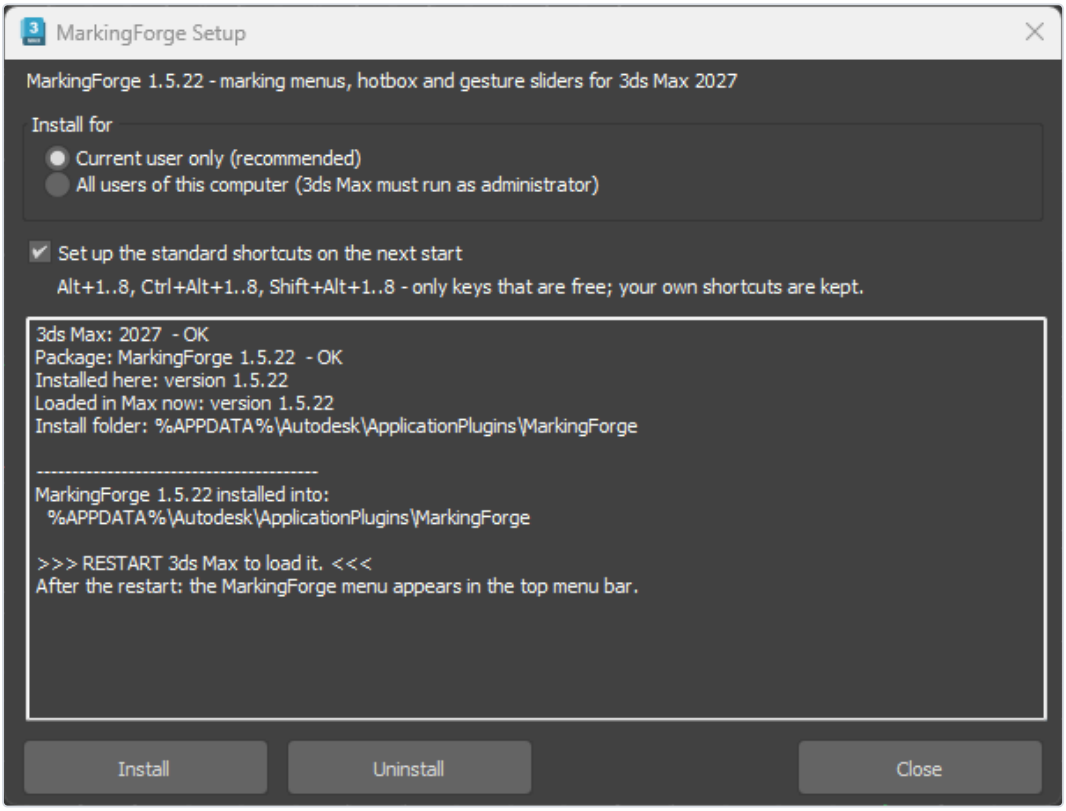
1. Close 3ds Max.
2. Open Explorer, paste `%APPDATA%\Autodesk\ApplicationPlugins` into the address bar and press Enter (create the `ApplicationPlugins` folder if it does not exist). For every user of the computer, use `%ProgramData%\Autodesk\ApplicationPlugins` instead - Windows asks for an administrator's permission.
3. Copy the folder `Manual install\MarkingForge` into it, with its name unchanged.
4. Start 3ds Max. The first start gives the dials their keys by itself (chapter 5).

### Method 4 - the studio script (silent)

1. Close 3ds Max on that computer - the script stops when 3ds Max is running.
2. In the unzipped download, run from a command prompt (as administrator for every user of the computer, the default scope):

```
powershell -ExecutionPolicy Bypass -File "Studio\Deploy-MarkingForge.ps1" -Scope AllUsers
powershell -ExecutionPolicy Bypass -File "Studio\Deploy-MarkingForge.ps1" -Scope CurrentUser
```

It copies `Manual install\MarkingForge`, checks every file by SHA-256 and ends with exit code 0 on success. `-DryRun` shows what it would do and changes nothing; `-NoShortcuts` leaves the user's keys alone; `-StudioMenus` gives the user a prepared dial layout. Every switch, the exit codes and log-on scripts are in the **Studio Deployment Guide**.




After Install: the Setup window says where the plugin went and asks for a restart.

## Setup window options

| Option | What it does |
|---|---|
| Current user only (recommended) | Installs into `%APPDATA%\Autodesk\ApplicationPlugins\MarkingForge`. No administrator rights needed. |
| All users of this computer (3ds Max must run as administrator) | Installs into `%ProgramData%\Autodesk\ApplicationPlugins\MarkingForge`. Start 3ds Max as administrator first. |
| Set up the standard shortcuts on the next start | Ticked (the default): on the next start every dial without a key gets its standard one - Alt+1..8, Ctrl+Alt+1..8, Shift+Alt+1..8 - even when you already have keys on some dials. A key you already use for something else is skipped, and no shortcut of yours is changed. Cleared: the next start leaves your keys alone. |
| Status box | 3ds Max version, the package, the version already installed in the chosen folder, the version loaded right now, the install folder - and a WARNING for every other copy of the plugin it finds. |

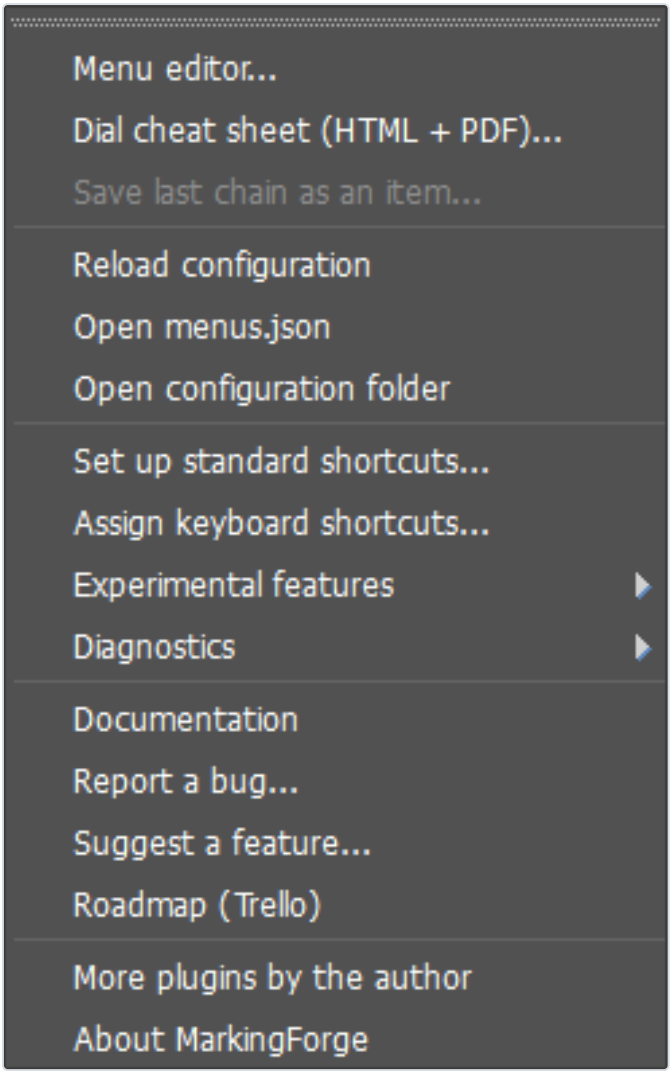
| | |
|---|---|
| Install / Update / Repair | Copies the plugin. The button says **Update / Repair** when MarkingForge is already in the chosen folder; it then replaces it, keeping your dials and settings. It works even while the old version is loaded. Greyed out while a check fails. |
| Uninstall | After a question, removes MarkingForge from both folders - current user and all users. Your own settings are kept (see chapter 13). |
| Close | Closes the window; nothing else happens. |

> **Why a restart?** MarkingForge is a C++ plugin, and 3ds Max loads C++ plugins only while it starts. The installer copies the files; the next start loads them.

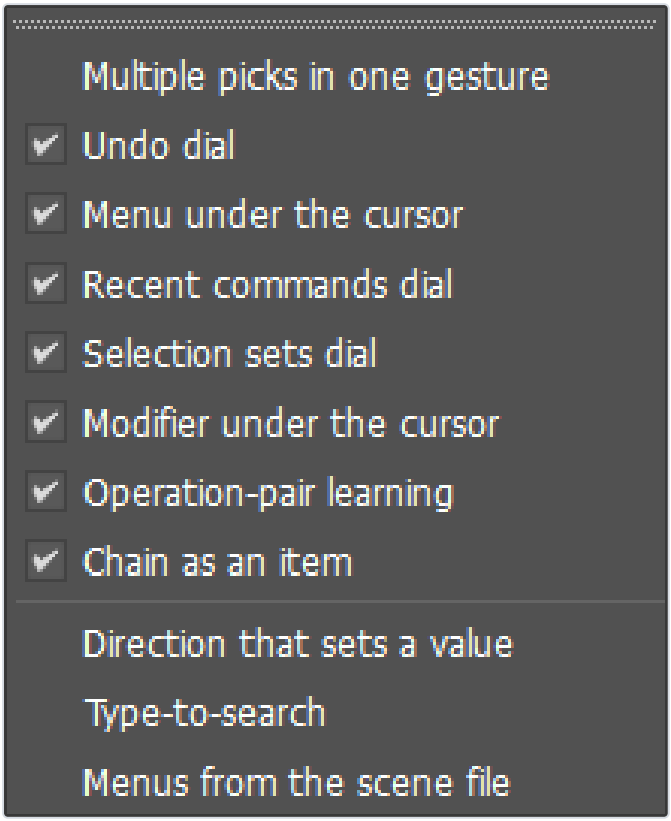
# 4. The first start and checking that it works

On the first start after installing, MarkingForge sets itself up - whichever method you used:

- a **MarkingForge** menu appears in the menu bar, before *Help*;
- its settings folder is created, with `menus.json` holding the 24 built-in dials (chapter 12);
- the dials get their keys, Alt+1..8, Ctrl+Alt+1..8 and Shift+Alt+1..8, unless something else already uses them (chapter 5). The MAXScript Listener says how many, for example *MarkingForge: standard shortcuts set up for 24 dial(s), 0 skipped (key in use)*.



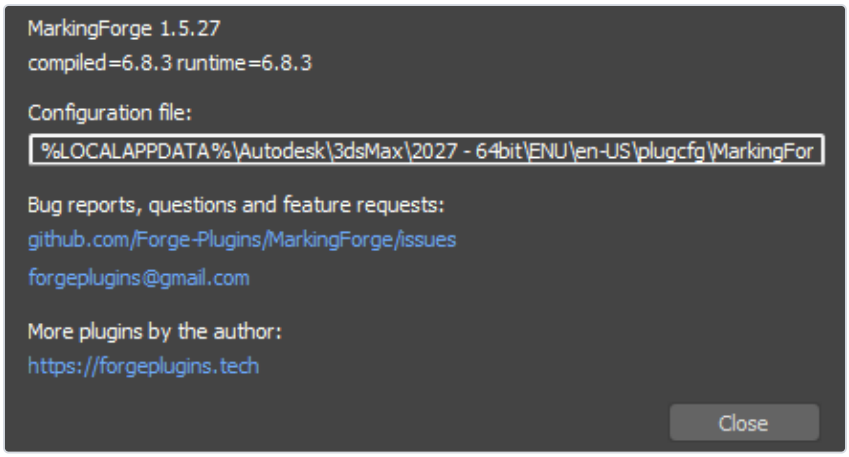

The MarkingForge menu.




Its Experimental features submenu - options you switch on deliberately.

### Four checks

1. Hold **Alt+1** with the cursor over a viewport: the Modelling dial appears. Release in the centre to cancel.
2. **MarkingForge > About MarkingForge** shows the version and the configuration file in use.
3. **MarkingForge > Menu editor...** opens the editor; its title bar names both versions - *MarkingForge - menu editor 0.20.40 | plugin 1.5.108*.
4. In the MAXScript Listener (F11) type `MarkingForge.version()`. It prints the version, 1.5.108, followed by a short description. "Unknown property" means the plugin was not loaded - see Troubleshooting.

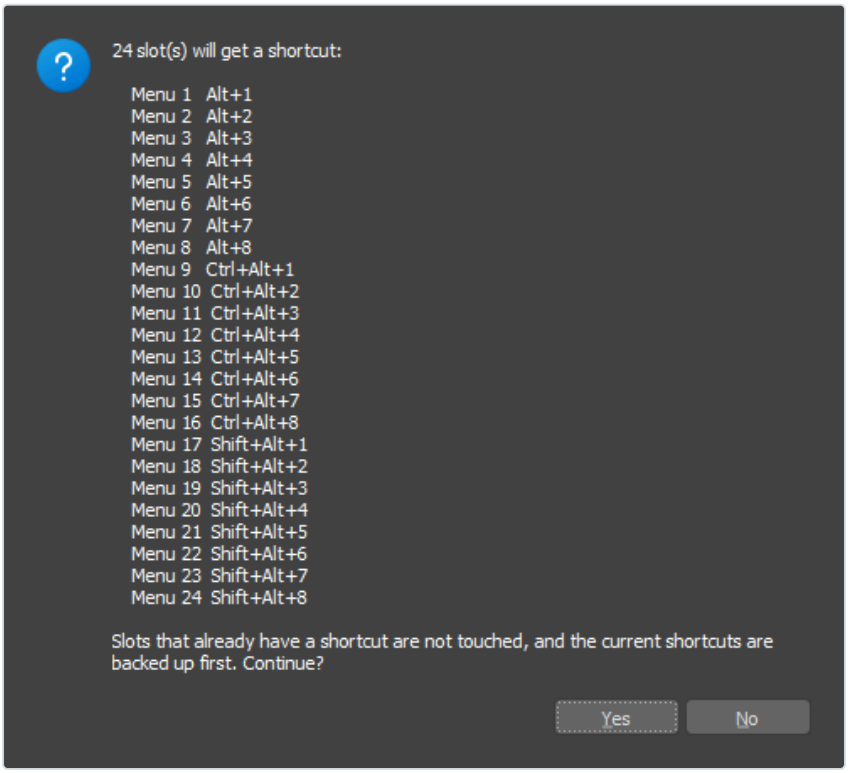


About MarkingForge: the version that is loaded, the configuration file and where to write.

# 5. Keyboard shortcuts

Each of the 24 dials opens by holding a key, and the keys are there from the first start: when no dial has a key yet, the first start gives each its standard one - whatever way MarkingForge was installed - skipping keys something else uses and changing no shortcut of yours. This happens once per 3ds Max version and Windows user. The installer's tick box *Set up the standard shortcuts on the next start* also fills the free dials of somebody who already has some keys; cleared, it tells the first start to leave the keys alone (the studio script's `-SetupShortcuts` and `-NoShortcuts` do the same).

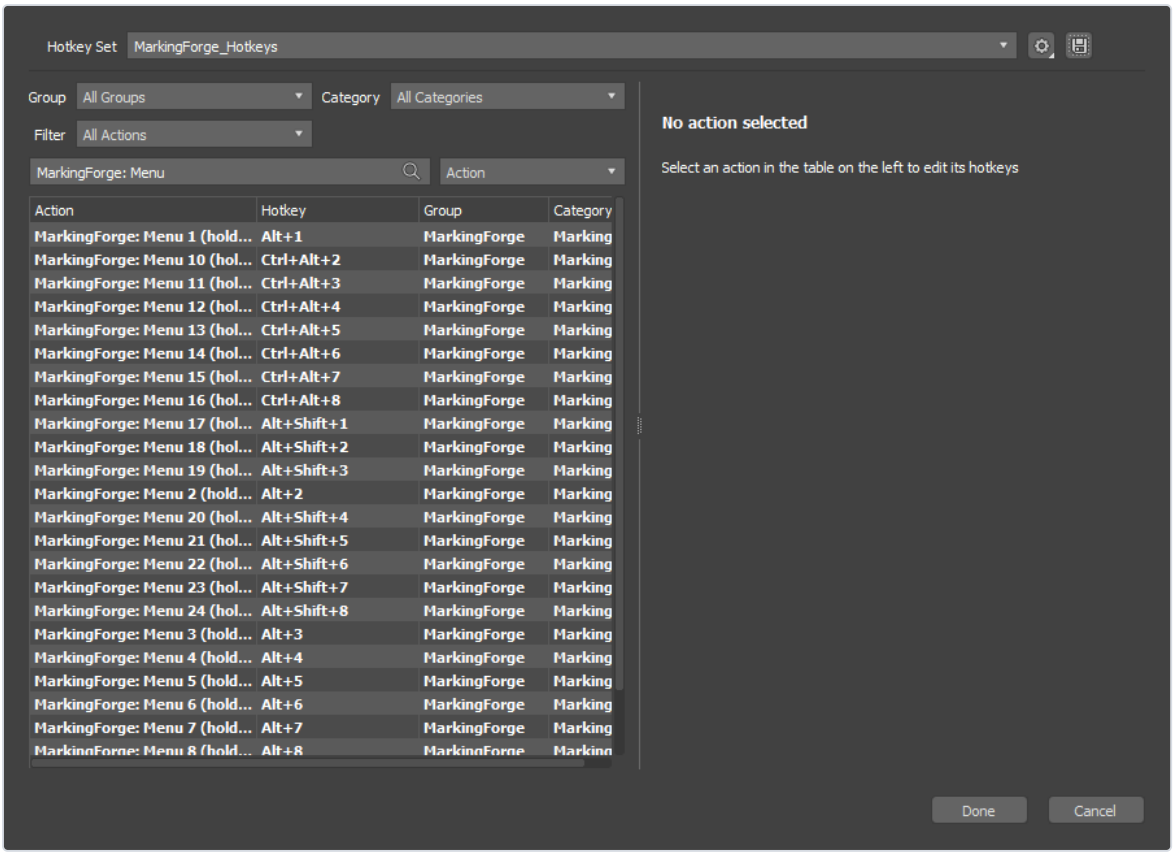
To fill in dials that are still without a key at any later time, choose **MarkingForge > Set up standard shortcuts...** It shows exactly what it will do and asks once:




Set up standard shortcuts: only dials without a shortcut, only keys nothing else uses.

| Key | Dial | Key | Dial | Key | Dial |
|---|---|---|---|---|---|
| **Alt+1** | Modelling | **Ctrl+Alt+1** | Transform | **Shift+Alt+1** | Viewport display |
| **Alt+2** | Modifiers | **Ctrl+Alt+2** | Align and pivot | **Shift+Alt+2** | Selection sets |
| **Alt+3** | Selection | **Ctrl+Alt+3** | Lights and cameras | **Shift+Alt+3** | Materials |
| **Alt+4** | Create | **Ctrl+Alt+4** | UV mapping | **Shift+Alt+4** | Render |
| **Alt+5** | Viewport shading | **Ctrl+Alt+5** | Snaps and grid | **Shift+Alt+5** | Animation |
| **Alt+6** | Modifier stack | **Ctrl+Alt+6** | Show and hide | **Shift+Alt+6** | Scene |
| **Alt+7** | Undo | **Ctrl+Alt+7** | View | **Shift+Alt+7** | Tools and setup |
| **Alt+8** | Recent commands | **Ctrl+Alt+8** | Viewports | **Shift+Alt+8** | File |

You can change any of them in the menu editor (**This slot's shortcut:** - **Set shortcut...**, then **Apply shortcuts**; **Fill the free slots...** proposes the standard scheme) or in 3ds Max's own **Customize > Hotkey Editor** (also **MarkingForge > Assign keyboard shortcuts...**), where they are listed as *MarkingForge: Menu 1 (hold = marking menu)* ... *Menu 24*. The same category holds *Menu 1 - next set* ... *Menu 24 - next set* for dials with several sets.



The Hotkey Editor filtered to MarkingForge. Any key can be reassigned here as well.

> MarkingForge writes shortcuts into a hotkey set of its own, **MarkingForge_Hotkeys**, which 3ds Max then shows as the active Hotkey Set. It holds ALL your shortcuts - 3ds Max's and other plugins' as well as the dials' - and a backup is made before every change (chapter 12).

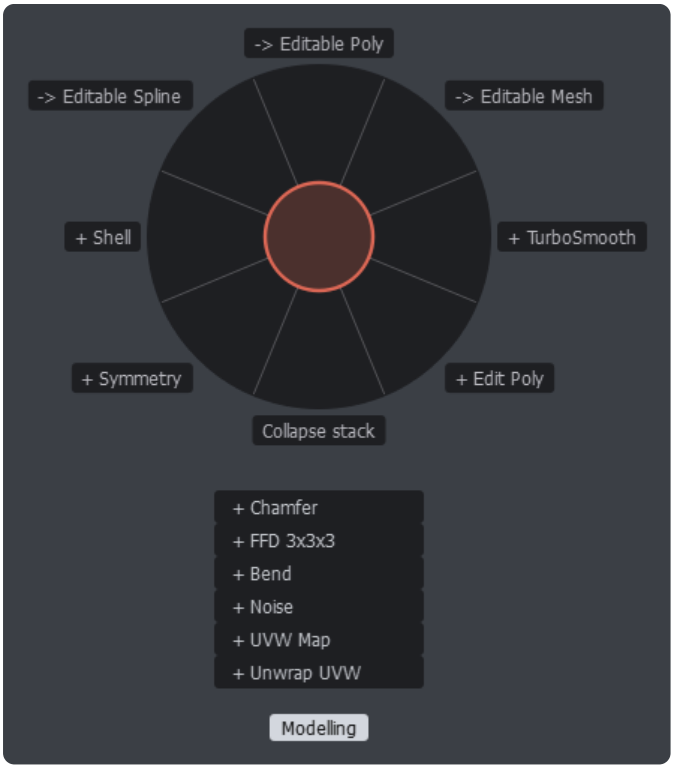
# 6. Using the dials

The same key does three things, depending on the hand:

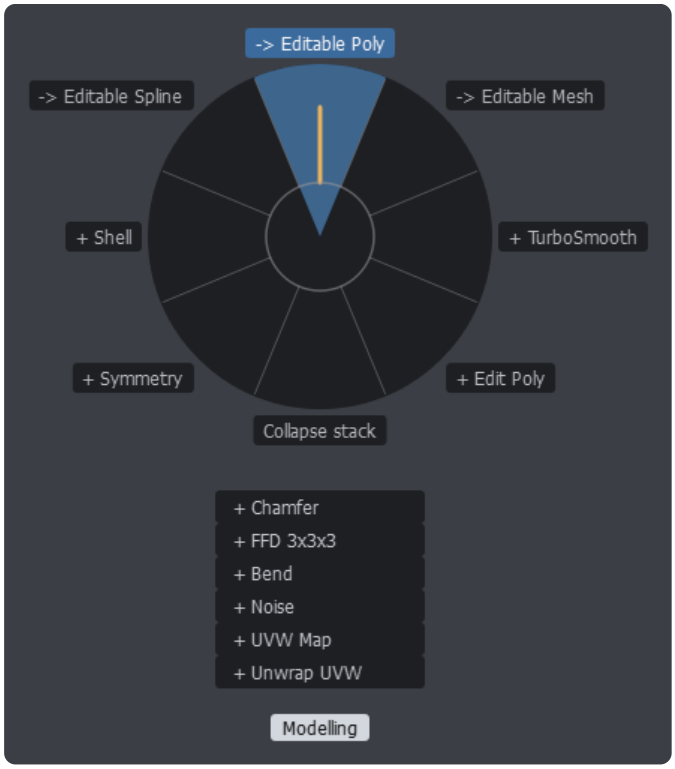
| Move | What happens |
|---|---|
| **Hold** | Hold the key: the dial appears around the cursor. Move towards an item and release the key. For learning and browsing. |
| **Flick** | Press the key and move the mouse at once, without waiting. The item runs without the dial being drawn. After a week or two you stop looking - the direction is in your hand. |
| **Tap** | Press and release without moving. Does nothing by default; the editor can give it one of the directions, and *Experimental features* > *Tap repeats the last modelling command* makes it run your last poly, mesh or spline command again (1.5.85). |

Releasing in the centre cancels. So do **Esc** and the right mouse button.

> **Aim with the direction - you do not have to hit the caption.** A dial picks by **direction**, not by the caption tile under the cursor. Move towards a command and release anywhere in its slice - near the centre, on its caption, or far beyond it, even outside the dial: the command runs. Only the **centre** cancels. Far out, three things differ: past the dotted ring a command with a settings window opens it (unless *No settings windows* is ticked), a submenu (>) opens, and a row of the list under the dial wins while the cursor is on it.




Alt+1 on an ordinary object: convert and add modifiers. Less common items sit in a list under the ring - rows are clicked, directions are aimed at.



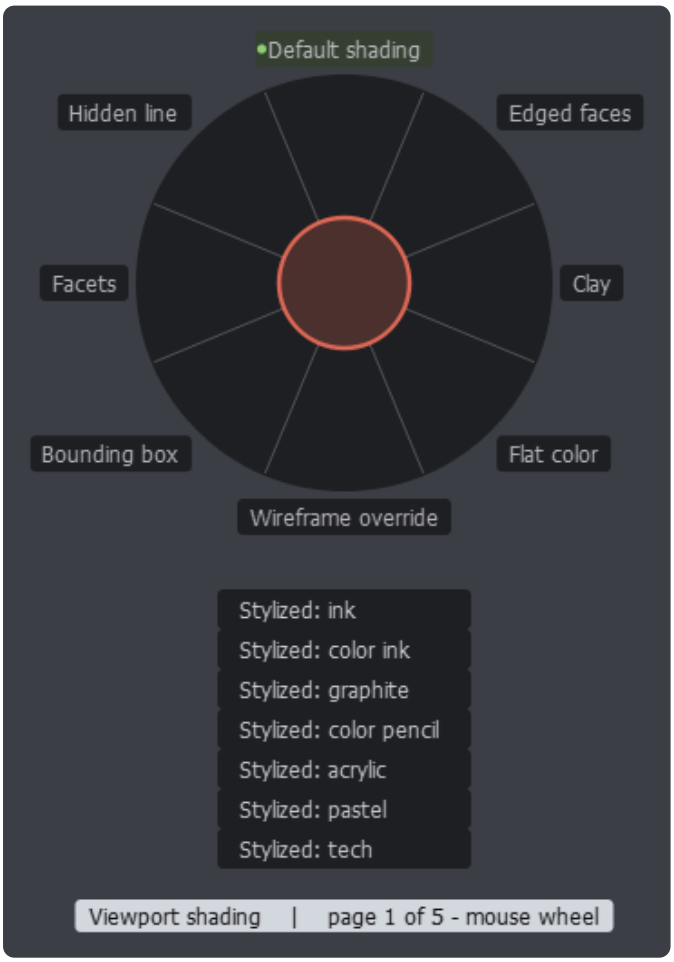

Aiming north: the direction lights up. Releasing runs *Convert to Editable Poly*.

Commands with a settings dialog (the caddy) - Chamfer, Extrude, Inset... - run at once on an ordinary release, like their button in the Modify panel. Moving **far past the ring** before releasing opens the dialog first, like the settings icon in a quad menu - "far" is the **dotted circle** outside the dial, which brightens once you are past it. A flick counts the same way: longer than the distance to that circle, the dialog opens. Either can be made the default per command (**Settings...** in the editor). How far out the circle lies - 20 px past the farthest caption by default, 0 to 200 px - is set with **Settings ring...** in the editor (1.5.10).

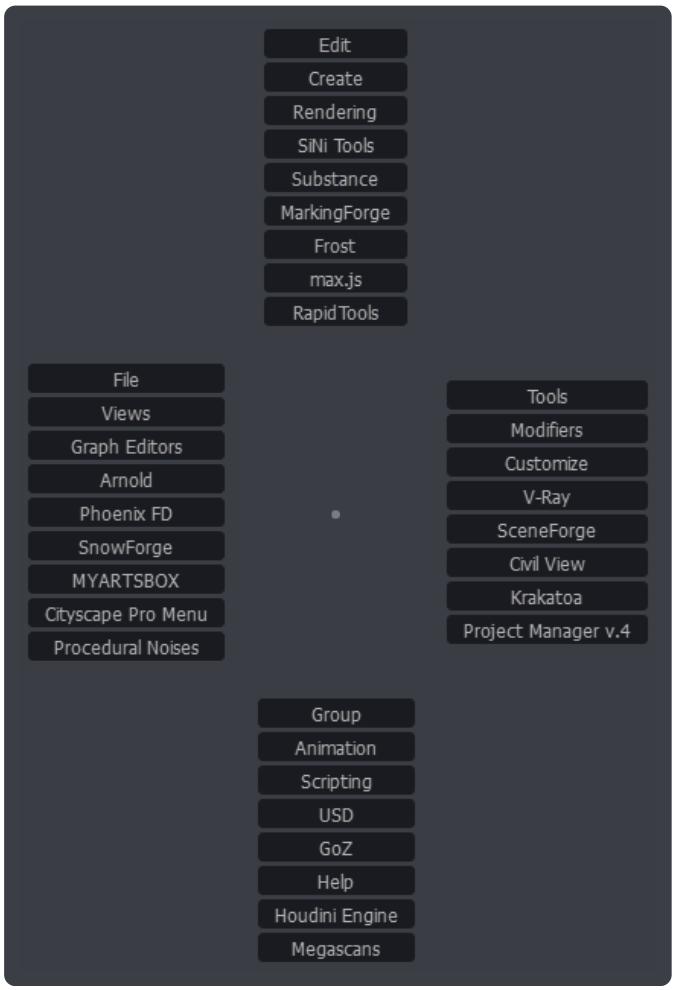
More tricks - several commands in one gesture, the dial of the object under the cursor, the suggested next command and search by typing - are in the **QuickStart Long** ("Five tricks worth knowing") and in chapter 2 of the **Reference**.

**Alt+5** drives the **viewport**, on four pages: how it is *drawn* (default shading, clay, facets, flat colour, hidden line, bounding box, wireframe override, edged faces - the stylized looks in the list), how it is *lit* (shadows, ambient occlusion, highlights, scene or default lights, materials with or without maps), what it *shows* (grid, safe frames, statistics, ViewCube, selection display, isolate - hiding whole categories in the list) and where it is *seen from* (top, front, left, right, perspective, orthographic, camera, maximize - zooms, layouts and the field of view in the list). Hold Alt+5 and turn the mouse wheel towards you for the next page. A switch that is a 3ds Max command is drawn checked while it is on.

One wheel notch **away** from you, Alt+5 is the **hotbox**: all of 3ds Max's main menu as tiles around the cursor, including menus other plugins add - point at a menu title, then at an entry, and release; or click the entry with the left mouse button, as in Maya (a right click or Escape closes the hotbox).

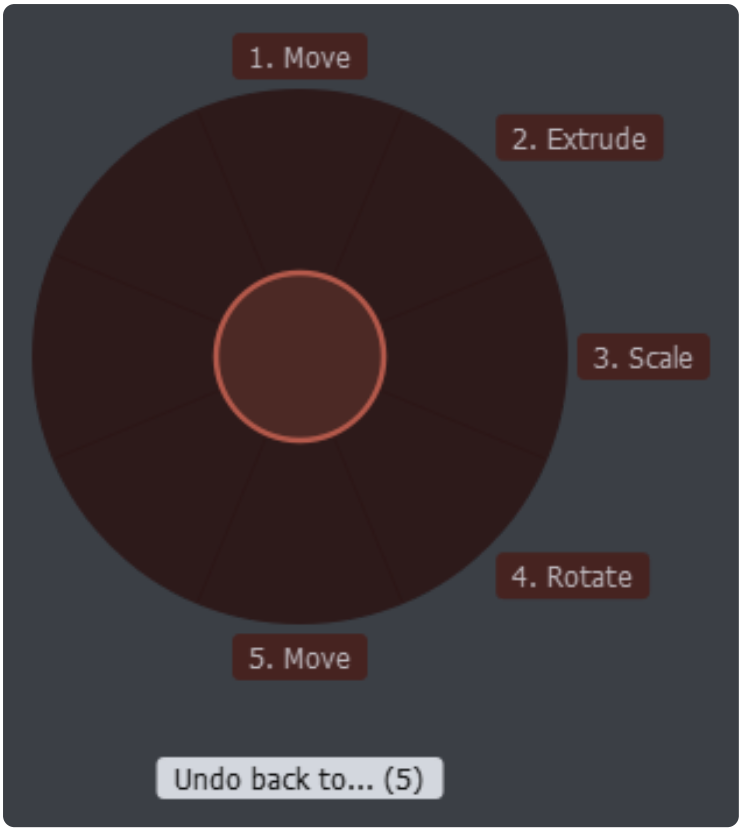



Alt+5 - the first page: how the viewport is drawn.




Alt+5, one wheel notch away - the hotbox.

**Alt+7** is the **undo dial**: the last eight operations by name; N undoes one step, NE two, and so on.

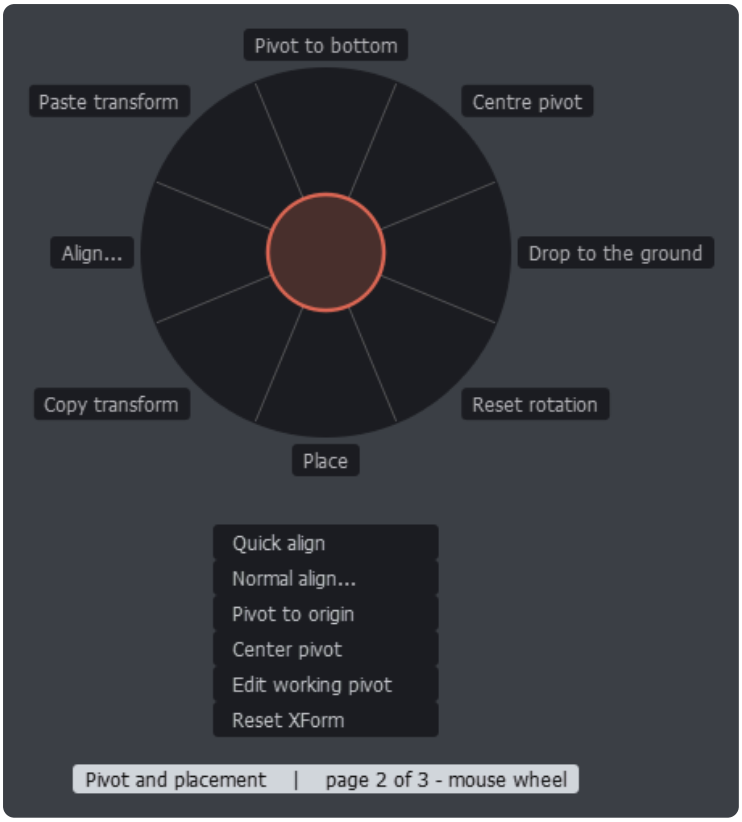



Alt+7 - the undo dial.

### Pages - the mouse wheel

One key can hold several dials. A dial with **sets** (made in the menu editor, or added from the dial library) has pages: hold its key and turn the **mouse wheel** - towards you for the next set, away for the previous one - and the dial shows it at the same place; release over a command to run it. The footer says where you are, *page 2 of 3 - mouse wheel*. The next press opens the page you turned to last - also after 3ds Max restarts.

The wheel is taken only while it can turn a page - on a dial with more than one set, at its first level, or in a hotbox that is one of those sets (the last page of Alt+5). In a submenu, during a search and on a dial with a single set (every built-in dial but Alt+5) it goes to 3ds Max as always, so zooming and scrolling are unchanged.



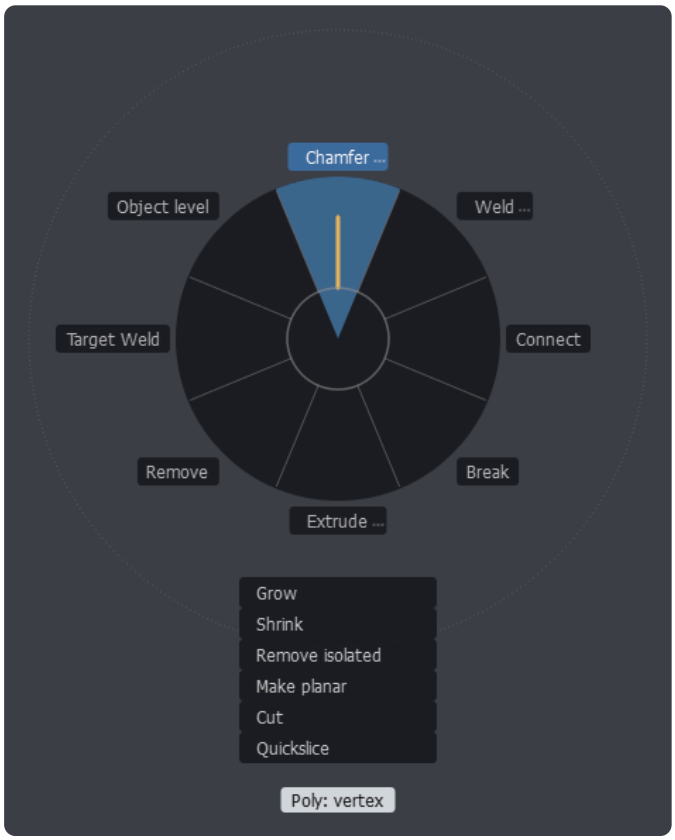
A dial with three sets, turned to its second page with the mouse wheel - see the footer.

# 7. Dials that follow what you are doing

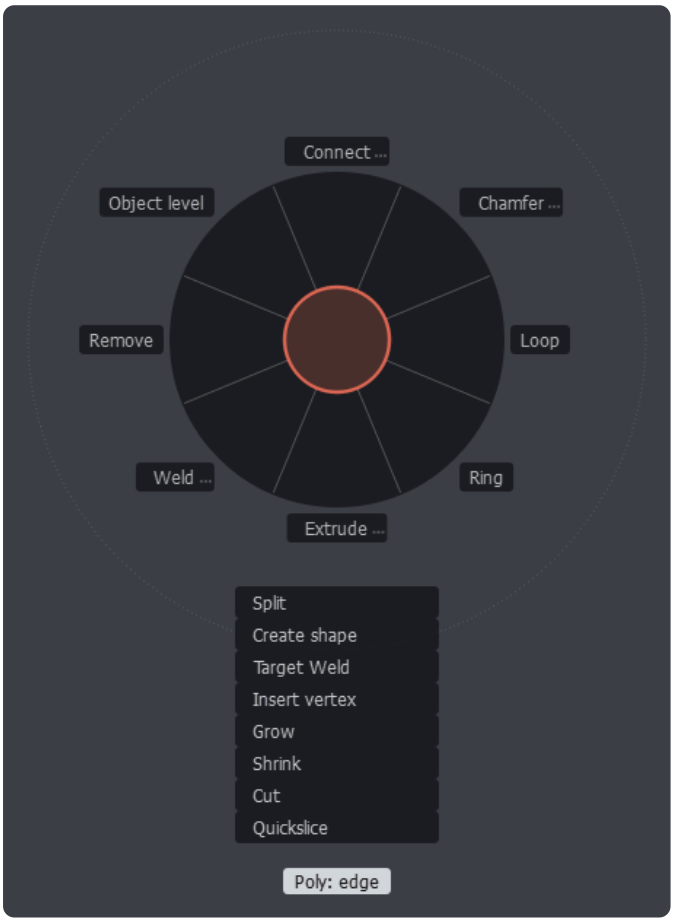
A dial can change its contents with the situation. The built-in **Modelling** dial (Alt+1) is the clearest example: *Convert to Editable Poly* and everything you do next live under the same key.




Alt+1 on an Editable Poly at object level: the sub-object levels, Attach, TurboSmooth, Collapse.




The same key at Vertex level: Chamfer, Weld, Connect, Break... NW always returns to object level.




Edge level: Connect, Chamfer, Loop and Ring side by side.

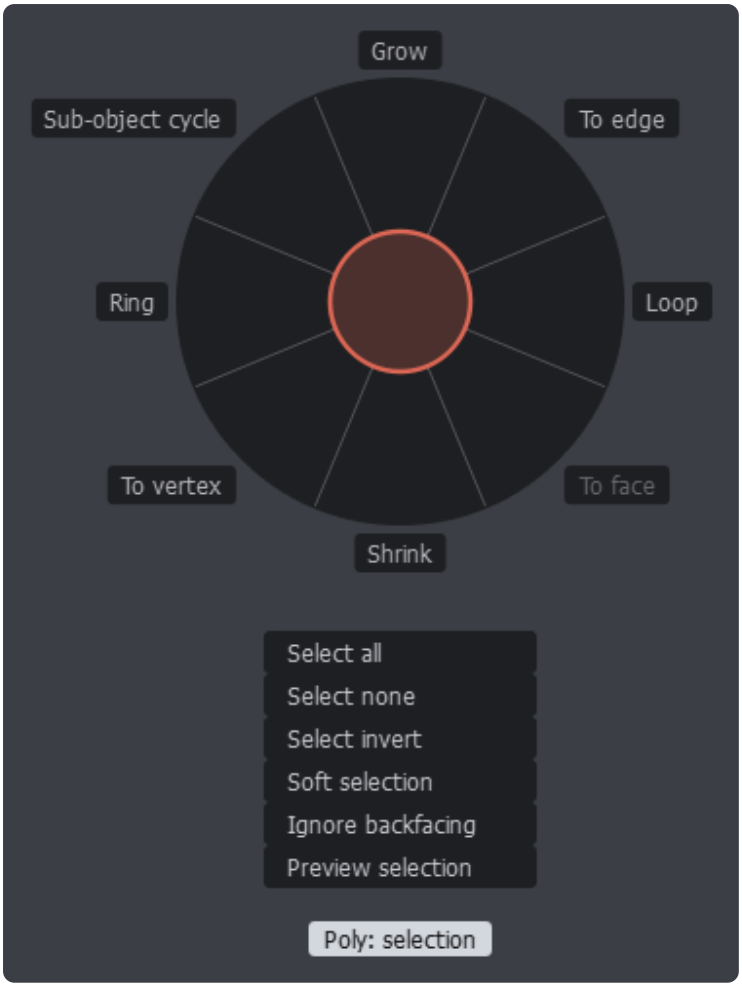



Polygon level, aiming at Inset.

The **same operation keeps the same direction** on an Editable Poly and under an Edit Poly modifier, so your hand does not have to know which one it is working on:

| Level | N | NE | E | SE | S | SW | W | NW |
|---|---|---|---|---|---|---|---|---|
| **Object** | Vertex | Edge | Border | Polygon | Element | Attach | + TurboSmooth | Collapse stack |
| **Vertex** | Chamfer | Weld | Connect | Break | Extrude | Remove | Target Weld | Object level |
| **Edge** | Connect | Chamfer | Loop | Ring | Extrude | Weld | Remove | Object level |
| **Border** | Cap | Chamfer | Extrude | Loop | Create shape | Shrink | Grow | Object level |
| **Polygon** | Extrude | Bevel | Inset | Outline | Hinge | Flip | Detach | Object level |
| **Element** | Detach | Hide | Flip | Unhide all | Hide unselected | Shrink | Grow | Object level |

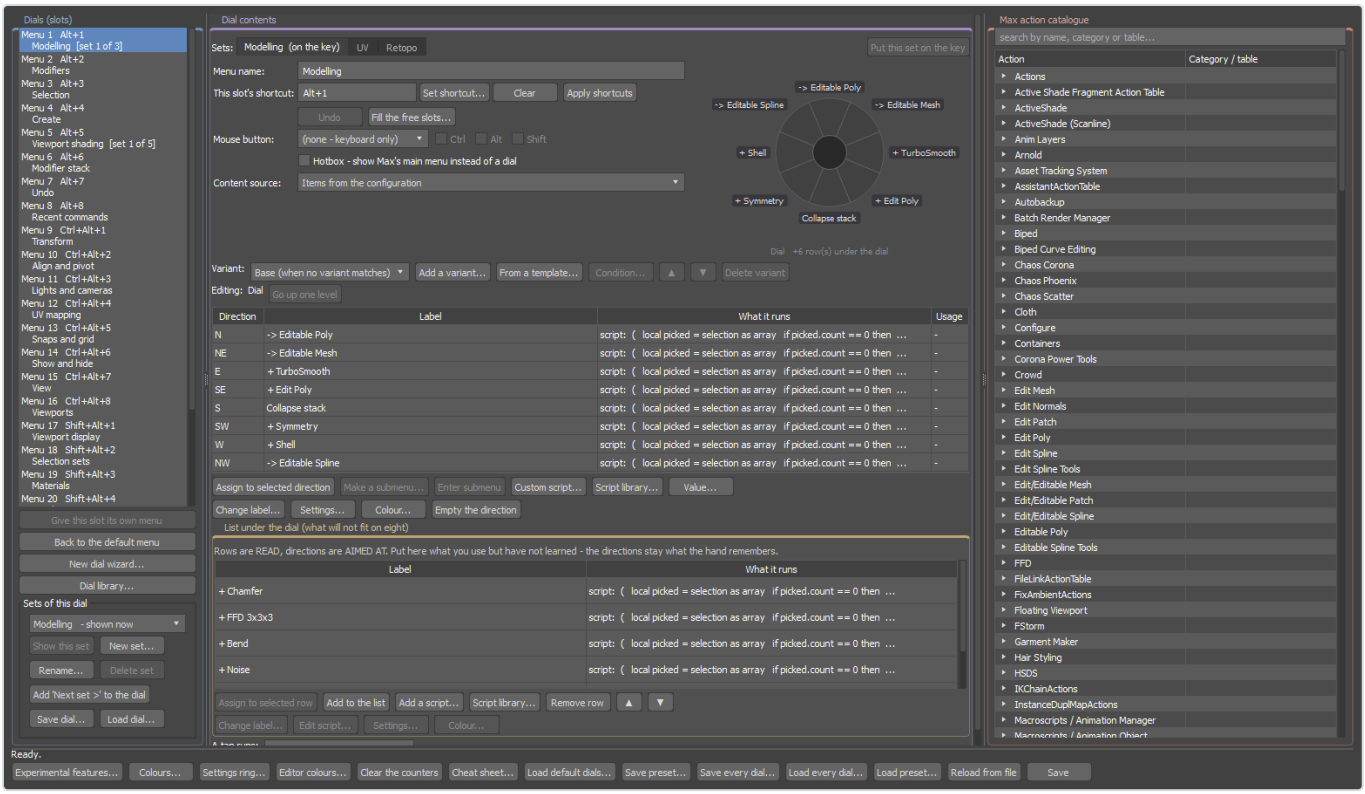
Other dials do the same: Selection offers Grow, Shrink, Loop and Ring at sub-object level; Create and Modifiers turn into shapes and shape modifiers when a shape is selected; Lights and cameras turns into camera or light controls; UV mapping follows an Unwrap UVW modifier; Materials, Animation and Scene change when the cursor is over the Slate editor, Track View or the Scene Explorer. Chapter 11 lists every case.



Alt+3 on an Editable Poly sub-object level: Grow and Shrink opposite each other, Loop and Ring opposite each other.

# 8. The menu editor

**MarkingForge > Menu editor...** Everything about the dials is set here; changes take effect as soon as you press **Save**, without restarting 3ds Max. This chapter is a map of the window - the **Editor Guide** takes you through every part of it step by step.




The menu editor: dials on the left, the selected dial's directions in the middle, 3ds Max's 4000+ commands on the right.

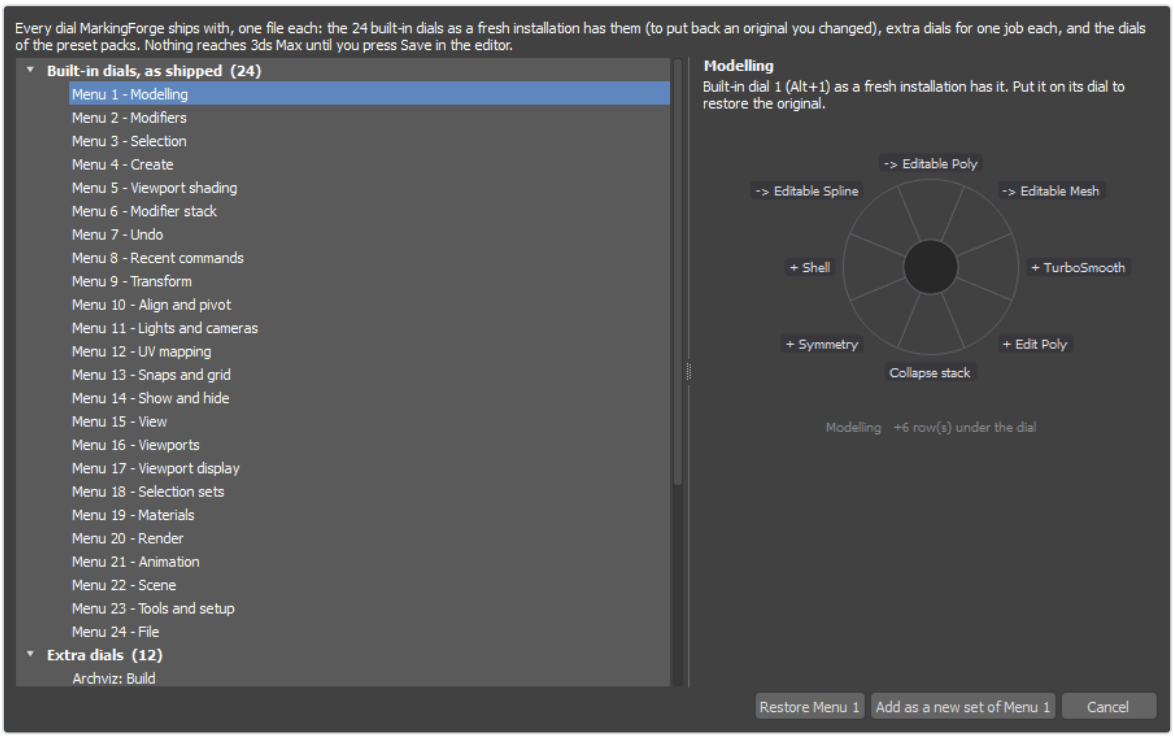
| Area | Use |
|---|---|
| Dials (slots) - left | The 24 slots with their shortcuts. Pick one to edit it. Below the list: **Give this slot its own menu**, **Back to the default menu**, **New dial wizard...**, **Dial library...** and the box **Sets of this dial** (Show this set, New set..., Rename..., Delete set, Add 'Next set >' to the dial, Save dial..., Load dial...). |
| Right-click a dial | On the list of dials or on the drawing of the dial: save it to a file, load a dial file into it, put a dial from the library on it, or save / load every dial. |
| Sets - tabs | A dial with more than one set shows its sets as tabs over its contents; the one the key opens is marked *(on the key)*. A tab is looked at and edited without changing what the key opens - **Put this set on the key** does that. |
| Shortcut and mouse button | **This slot's shortcut:** Set shortcut..., Clear, Apply shortcuts, Undo, Fill the free slots... - and **Mouse button:** a button with at least one modifier key (for example Ctrl+Shift+Middle). |
| Content source | Items from the configuration, or a live source: modifier stack, undo list, recent commands, selection sets. |
| Variant | The contextual variants of the dial and their conditions (object class, sub-object level, modifier, window under the cursor, selection count, name, layer) - Add a variant..., From a template..., Condition... |
| Directions | N ... NW, and the dial drawn beside them (click a wedge to select it). Drag a command from the catalogue onto a row, or double-click a command. Below: Make a submenu..., Custom script..., Script library... (102 ready-made scripts), Value... (a gesture slider), Change label..., Settings..., Colour..., Empty the direction. |
| List under the dial | Up to twelve extra rows for what does not fit in eight directions. |
| Max action catalogue - right | Every command 3ds Max knows, including other plugins' - searchable by name. |

| | |
|---|---|
| Bottom row | Experimental features..., Colours..., Editor colours..., Clear the counters, Cheat sheet... (a printable PDF of your saved dials), Load default dials..., Save preset..., Save every dial..., Load every dial..., Load preset..., Reload from file, Save. The buttons wrap to more rows when the window is narrow. |

> To get the built-in layout back at any time: **Load default dials...** By default it fills only the dials that have no menu of their own. One dial alone: **Dial library...** - the built-in dial of the slot is already selected, and **Restore Menu N** puts the original back.

## The dial library

Every dial MarkingForge ships is also a file of its own (.mfdial): the 24 built-in dials as a fresh installation has them, 12 extra dials for one job each (mesh cleanup, pivots, cloning, smoothing, retopology, quick look, rigging, archviz) and the 15 dials of the preset packs. They are installed in the plugin's `dials` folder and lie in `Dials` in the download. **Dial library...** puts one on the slot (**Put on Menu N**) or adds it as a page of it (**Add as a new set of Menu N**) - adding a set never changes what the key opens.

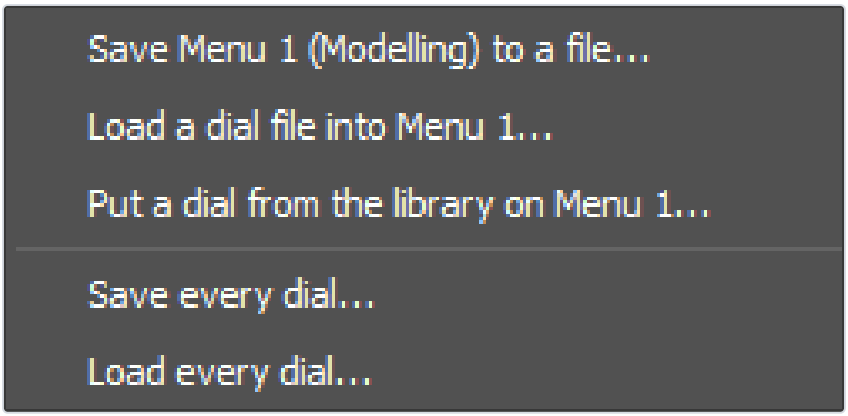



Dial library...: the built-in, extra and pack dials, with a drawing of the one selected.

## Keeping your dials as files

| Button | What it does |
|---|---|
| Save dial... / Load dial... | One dial in a .mfdial file - to keep it or to send it. Loading puts it in place of what the slot shows or adds it as a new set. |
| Save every dial... | Writes each of the 24 dials - and every other set of a dial - to a file of its own, in a new dated folder. |
| Load every dial... | Puts a whole folder made by Save every dial... back: what each dial showed, its sets in their order, and the colours of every dial. Dials the folder does not hold stay as they are; a damaged file stops the whole load; nothing is written until **Save**. |

Scripts in dial files are somebody else's code: loading asks, and removes them unless you answer that you trust the files.



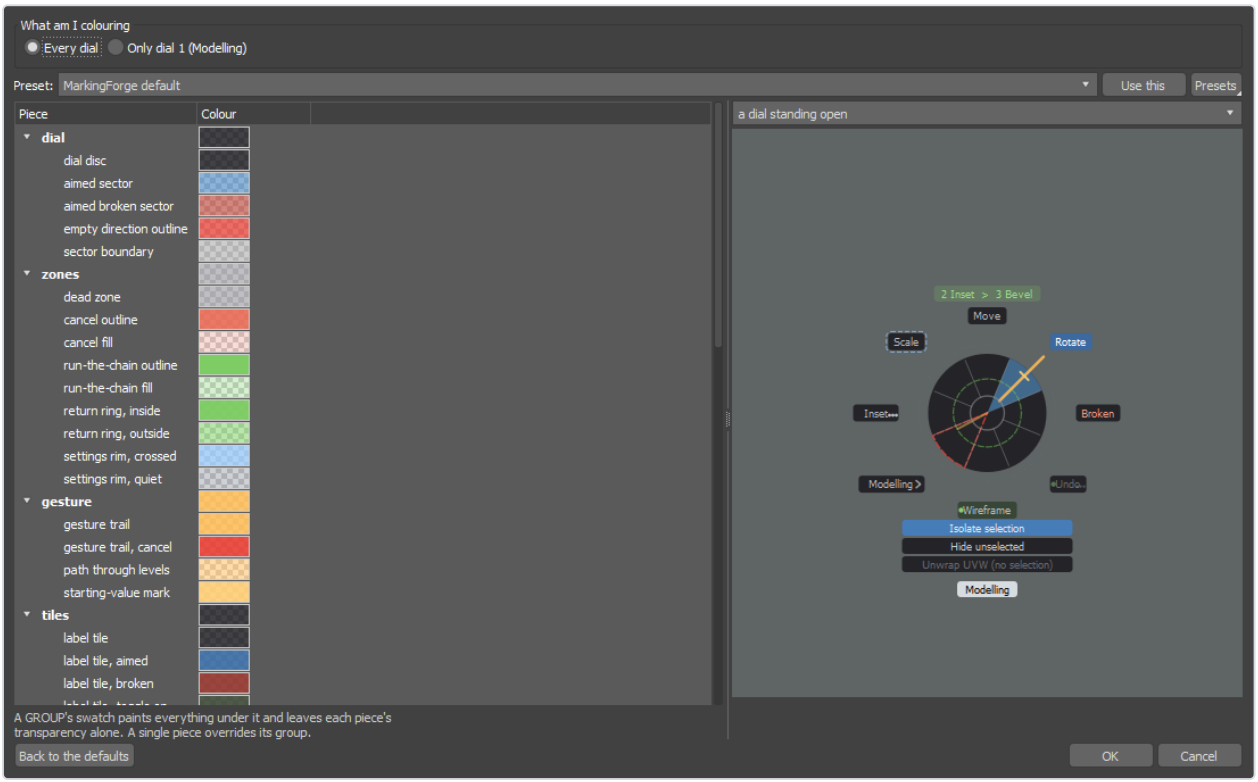

Right-click a dial - in the list or on its drawing - for the same commands where the dial is.

Closing the editor with unsaved changes (Esc or the window's X) asks whether to **Save**, **Discard** or **Cancel** - work is never thrown away without a question. If menus.json changed on disk while the editor was open, Save does not overwrite it: it offers **Save your edits as a separate copy**.

# 9. Colours, mouse buttons and experimental features

## Colours of the dials

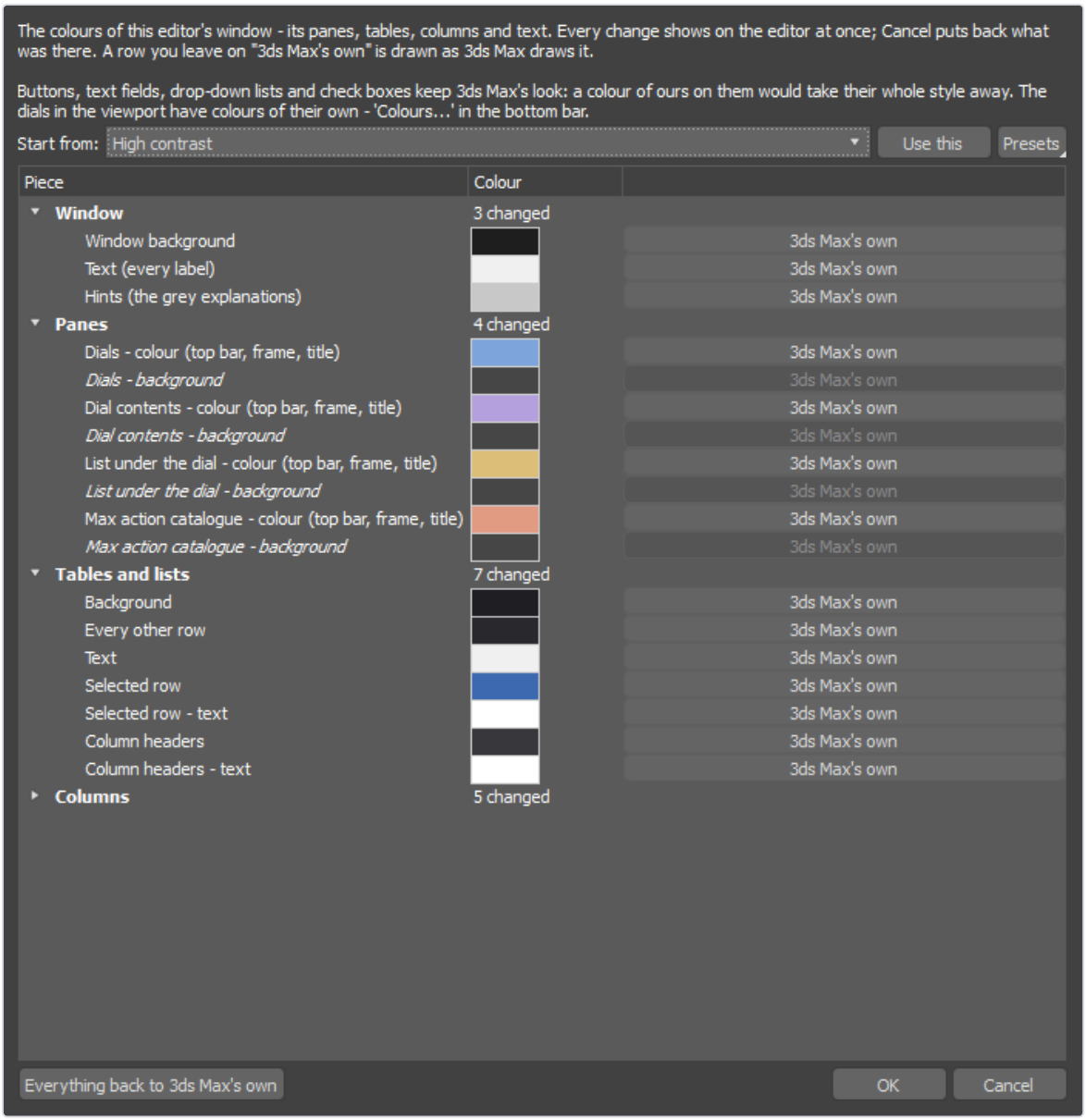
**Colours...** in the editor sets the palette of the dials. *What am I colouring* chooses **Every dial** or **Only** the dial you are editing - and, for a dial with sets, every page of it or one page alone (the pages the mouse wheel turns). The **Preset:** row at the top offers 31 ready-made looks; **Use this** paints the chosen one, and the **Presets** button saves your own (*Save these colours as a preset...*), deletes it, and exports or imports one preset as a `.mfcolors` file for another computer. **Colour...** under the directions colours a single item. The preview shows the dial in the moment you choose: standing open, aiming at a direction, about to cancel, with a chain waiting.




The colour editor, with its presets at the top.

## The editor's own colours

**Editor colours...** colours the editor's window itself - not the dials: its ground, text and hints, each pane, the tables and lists, every column. The window changes as you pick, and Cancel puts back what was there. It has 22 ready-made looks (the first is 3ds Max's own) and your own presets the same way as the dials. These colours are kept in `editor_colors.json` beside menus.json; they are not part of your dials and need no Save.



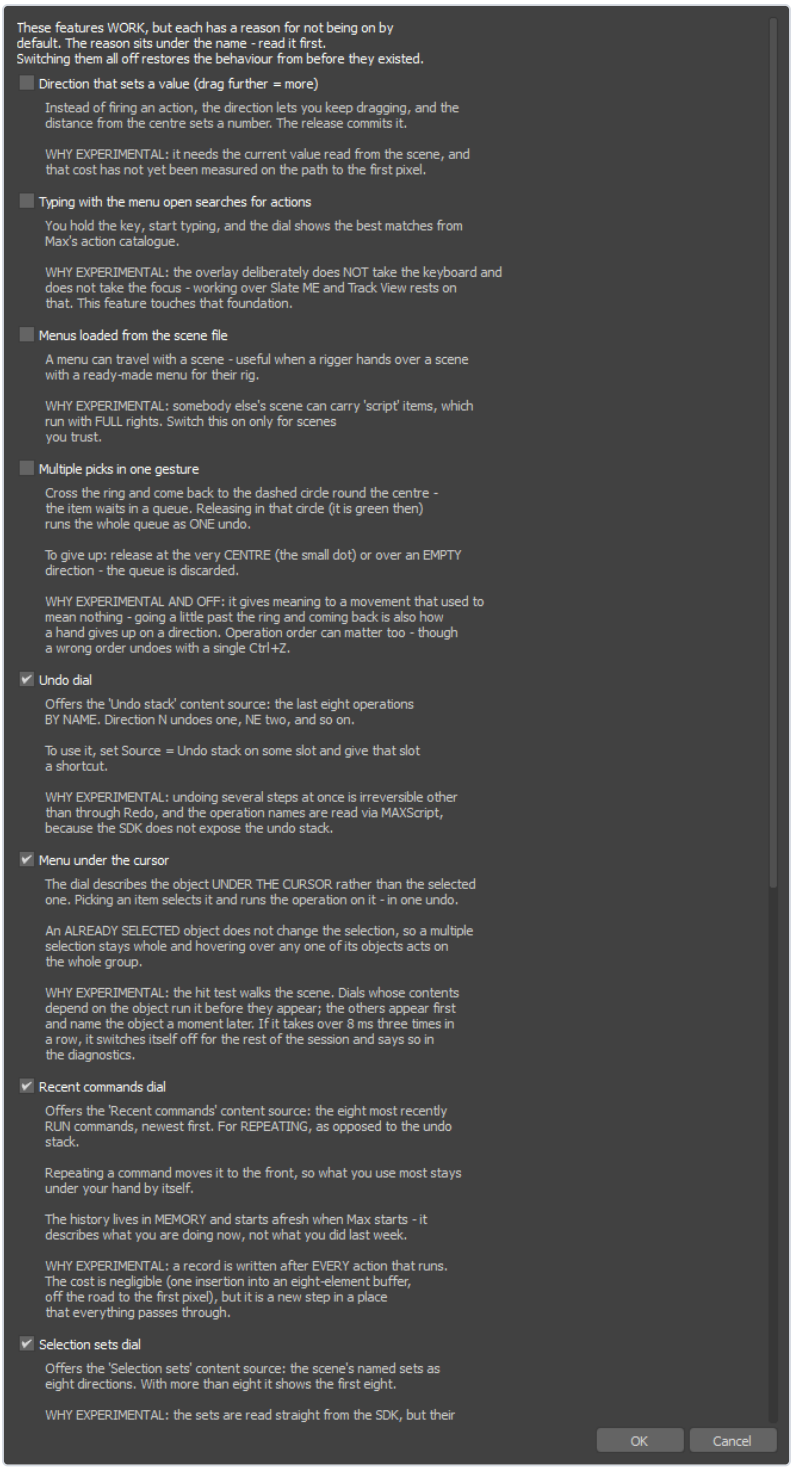

Editor colours...: the menu editor's own window in one of the ready-made looks.

## Opening a dial with a mouse button

Next to the shortcut, pick a mouse button and at least one modifier key. A bare button is refused on purpose: left selects, middle pans and right opens the quad menu - taking one of those away would break 3ds Max.

## Experimental features

Options that change how gestures behave - several picks in one gesture, the menu of the object under the cursor, type-to-search, gesture sliders. Switch them in **Experimental features...** in the editor or in the MarkingForge menu.



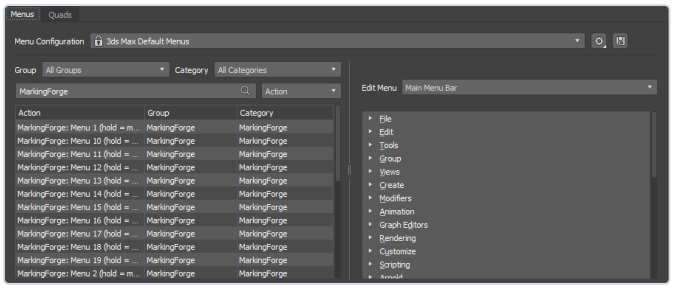
The experimental features, each with the reason it is optional.

# 10. Putting MarkingForge into the 3ds Max interface

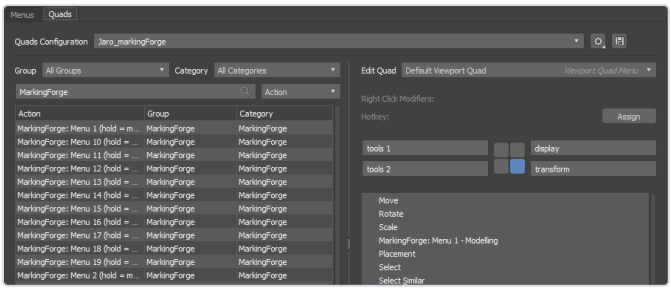
Every MarkingForge command is a standard 3ds Max action in the category **MarkingForge**. You can put any of them on a toolbar, into a menu or a quad menu, or give it a hotkey - exactly like 3ds Max's own commands.

## Into a menu or a quad menu

1. Choose **Customize > Menu Editor**.
2. On the **Menus** tab (or **Quads** for the right-click quad menu) type `MarkingForge` in the action search field.
3. Drag an action from the list into the menu tree on the right.
4. Close the editor with **Done**.



Menu Editor - MarkingForge actions ready to drag into a menu.



The same actions on the Quads tab, for the right-click quad menu.

## Onto a toolbar

Choose **Customize > Customize User Interface**, open the **Toolbars** tab, pick the category **MarkingForge** and drag an action onto any toolbar. A button shows the command's name; hovering it shows the description below.

## The commands

| Command | What it does |
|---|---|
| Menu editor... | Opens a window where you assign actions to dial directions and set keyboard shortcuts. |
| Dial cheat sheet (HTML + PDF)... | Collects every dial into one document to read and to print. Along the way it checks which dial would open empty. |
| Save last chain as an item... | Takes the last chain you ran (multiple picks) and stores it as ONE direction in a dial of your choice. |
| Reload configuration | Reads menus.json again. Useful when the file was edited outside Max. |
| Open menus.json | Opens the configuration file in your default text editor. The plugin picks up changes by itself. |
| Open configuration folder | Opens the folder holding menus.json, the shortcut backups and the settings in Explorer. |
| Assign keyboard shortcuts... | Opens 3ds Max's Hotkey Editor. The dials are the actions 'MarkingForge: Menu 1' ... 'Menu 24' - type MarkingForge into its search field. |
| Set up standard shortcuts... | Gives every dial without a shortcut its standard one: Alt+1..8, Ctrl+Alt+1..8, Shift+Alt+1..8. Keys already in use are skipped; your own shortcuts are not touched. |
| Configuration status | Shows which file is loaded, how many menus it holds, whether it has errors, and the state of the experimental features (including how often pair-learning suggestions were taken). |
| About MarkingForge | Shows the plugin's version, its configuration file, and links to support and to more plugins by the author. |
| Documentation | Opens the folder with the MarkingForge guides (QuickStart, Reference, Installation and more). |

| | |
|---|---|
| Report a bug... | Copies your 3ds Max and MarkingForge details to the clipboard and opens a window with links to the bug report form and the e-mail address. |
| Suggest a feature... | Send an idea or a feature request to forgeplugins@gmail.com. |
| Discord community | Opens the MarkingForge Discord channel in your web browser - questions, tips and news. |
| Roadmap (Trello) | Opens the MarkingForge features roadmap on Trello in your web browser - what is planned and in progress. |
| More plugins by the author | Opens https://forgeplugins.tech in your web browser. |
| Multiple picks in one gesture | Cross the ring and come back to the dashed circle - the item waits in a queue. Releasing in the circle runs the queue in order; Inset, Bevel... wait for a right-click. |
| Undo dial | Offers the "undo" menu source: the last eight operations by name. Assign it to a slot in the menu editor. |
| Menu under the cursor | The dial describes the object UNDER THE CURSOR rather than the selected one. Picking an item selects that object and runs the operation on it - in a single undo. |
| Direction that sets a value | A "value" direction turns into a slider - the gesture radius sets the number. |
| Type-to-search | Typing while the dial is open turns it into a search over Max's actions. |
| Menus from the scene file | CAUTION: menus carried in by somebody else's scene. Script items are rejected, but switch this on deliberately all the same. |
| Modifier under the cursor | A dial with the 'Modifier stack' source shows the stack of the object UNDER THE CURSOR rather than the selected one. Requires 'Menu under the cursor' - without it there is nowhere to get the object from. |
| Operation-pair learning | The plugin remembers what follows what and sets the dial's DEFAULT direction to the most frequent successor. It does NOT move items around - muscle memory rests on that. Session memory only. |
| Chain as an item | After you run a chain (multiple picks) the plugin remembers it. You then store it with 'Save last chain...' - with no dialog popping up after every gesture. |
| Maya marking menu look | Ticked: every dial looks like Maya's marking menu - no disc, grey boxes round a centre dot, a line to the cursor, Maya's option boxes for settings. Your own dial colours come back when you clear it. |
| No object outline in sub-object mode | Ticked: while you edit vertices, edges or polygons, 3ds Max's outline of the object under the cursor is switched off - over a mirrored half (Symmetry) it outlines the object BEHIND. It comes back when you leave the sub-object level. |
| Tap repeats the last modelling command | Ticked: a tap of a dial's key (press and release without moving) runs your last modelling command again - the last poly command on a poly object, the last spline command on a spline, the last mesh command on a mesh. |
| Chain submenus (Maya style) | Always on since 1.5.102: a submenu opens under the cursor, where your hand crossed the rim, and the centre of every menu you came through stays as a small circle - move onto one to go back to it. For one menu: the editor's Chain submenus row, Off. |
| No settings windows (Bevel, Extrude...) | Ticked: commands never open their settings window from a dial - no outer ring, no '...', every release runs the command at once, as its Modify-panel button does. |
| Recent commands dial | Offers the 'Recent commands' content source: the last eight you ran, newest first. For REPEATING, as opposed to the undo stack. |
| Selection sets dial | Offers the 'Selection sets' content source: the scene's named sets as eight directions. |
| Menu 1 - next set ... Menu 24 - next set | Switches that dial to its next set - one action per dial, for a key of its own. Sets are made in the menu editor. |

The 24 dials themselves are the actions *MarkingForge: Menu 1 ... Menu 24*. They are meant for keys and mouse buttons: a dial lives while the key is held, so a toolbar button cannot hold it open.

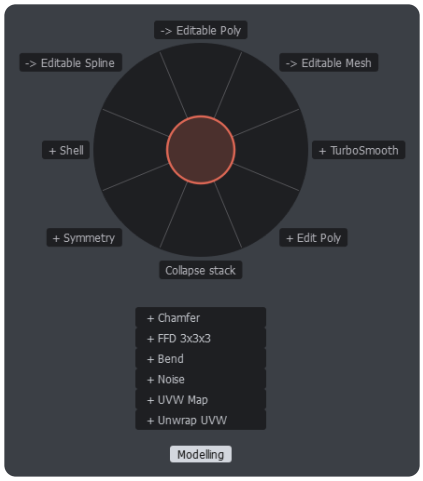
# 11. The 24 built-in dials

The layout a fresh installation starts with. Opposite directions hold opposite operations (All / None, Grow / Shrink, Hide / Unhide), so the hand learns one movement and its reverse. The grids show the directions as they sit on screen; the dial's name is in the centre.

## Alt+1 Modelling

dial 1

| | | |
|---|---|---|
| **NW**<br>-> Editable Spline | **N**<br>-> Editable Poly | **NE**<br>-> Editable Mesh |
| **W**<br>+ Shell | Modelling | **E**<br>+ TurboSmooth |
| **SW**<br>+ Symmetry | **S**<br>Collapse stack | **SE**<br>+ Edit Poly |




**List under the dial:** + Chamfer · + FFD 3x3x3 · + Bend · + Noise · + UVW Map · + Unwrap UVW

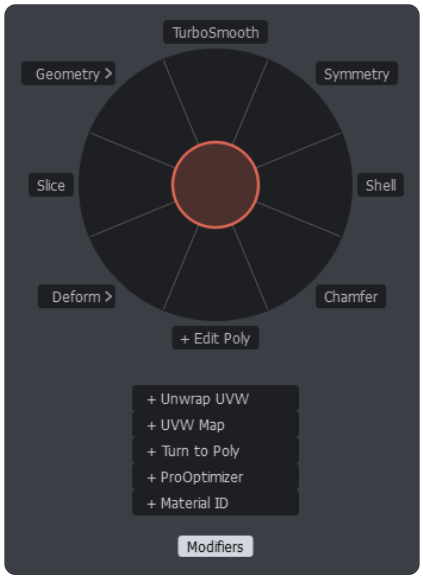
Changes with the situation (first match wins):

- **Materials (Slate)** - when cursor over the Slate Material Editor
- **Track View** - when cursor over Track View
- **Particle View** - when cursor over Particle View
- **Scene Explorer** - when cursor over the Scene Explorer
- **Modifier stack** - when cursor over the command panel
- **UV (Unwrap)** - when Unwrap UVW is current in the Modify panel
- **Edit Poly: vertex** - when Edit Poly is current in the Modify panel, sub-object level: Vertex
- **Edit Poly: edge** - when Edit Poly is current in the Modify panel, sub-object level: Edge
- **Edit Poly: border** - when Edit Poly is current in the Modify panel, sub-object level: Border / Segment
- **Edit Poly: polygon** - when Edit Poly is current in the Modify panel, sub-object level: Polygon
- **Edit Poly: element** - when Edit Poly is current in the Modify panel, sub-object level: Element
- **Edit Poly: object** - when Edit Poly is current in the Modify panel
- **Poly: vertex** - when the object is an Editable Poly, sub-object level: Vertex
- **Poly: edge** - when the object is an Editable Poly, sub-object level: Edge
- **Poly: border** - when the object is an Editable Poly, sub-object level: Border / Segment
- **Poly: polygon** - when the object is an Editable Poly, sub-object level: Polygon
- **Poly: element** - when the object is an Editable Poly, sub-object level: Element
- **Poly: object** - when the object is an Editable Poly
- **Mesh: vertex** - when EditMeshMod is current in the Modify panel, sub-object level: Vertex
- **Mesh: edge** - when EditMeshMod is current in the Modify panel, sub-object level: Edge
- **Mesh: face** - when EditMeshMod is current in the Modify panel, sub-object level: Border / Segment
- **Mesh: polygon** - when EditMeshMod is current in the Modify panel, sub-object level: Polygon
- **Mesh: element** - when EditMeshMod is current in the Modify panel, sub-object level: Element
- **Mesh: object** - when EditMeshMod is current in the Modify panel
- **Mesh: vertex** - when the object is an Editable Mesh, sub-object level: Vertex
- **Mesh: edge** - when the object is an Editable Mesh, sub-object level: Edge
- **Mesh: face** - when the object is an Editable Mesh, sub-object level: Border / Segment
- **Mesh: polygon** - when the object is an Editable Mesh, sub-object level: Polygon
- **Mesh: element** - when the object is an Editable Mesh, sub-object level: Element
- **Mesh: object** - when the object is an Editable Mesh
- **Spline: vertex** - when a shape (spline) is selected, sub-object level: Vertex
- **Spline: spline** - when a shape (spline) is selected, sub-object level: Spline
- **Spline** - when a shape (spline) is selected
- **Nothing selected** - when nothing is selected

## Alt+2 Modifiers

dial 2

| | | |
|---|---|---|
| **NW**<br>Geometry > | **N**<br>TurboSmooth | **NE**<br>Symmetry |
| **W**<br>Slice | **Modifiers** | **E**<br>Shell |
| **SW**<br>Deform > | **S**<br>+ Edit Poly | **SE**<br>Chamfer |




**Submenu Deform (SW):** N + Bend · NE + Twist · E + Taper · SE + Noise · S + FFD 3x3x3 · SW + Relax · W + Stretch · NW + Skew

**Submenu Geometry (NW):** N Mirror · NE Array · E Cap Holes · SE Vertex Weld · S Face Extrude · SW + Quadify Mesh · W OpenSubdiv · NW MeshSmooth

**List under the dial:** + Unwrap UVW · + UVW Map · + Turn to Poly · + ProOptimizer · + Material ID

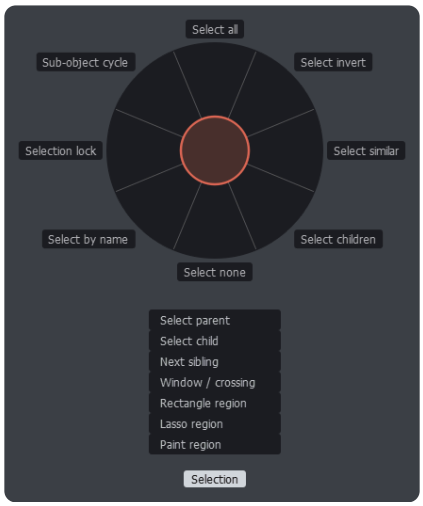
Changes with the situation (first match wins):

- **Shape modifiers** - when a shape (spline) is selected

## Alt+3 Selection

dial 3

| | | |
|---|---|---|
| **NW**<br>Sub-object cycle | **N**<br>Select all | **NE**<br>Select invert |
| **W**<br>Selection lock | **Selection** | **E**<br>Select similar |
| **SW**<br>Select by name | **S**<br>Select none | **SE**<br>Select children |




**List under the dial:** Select parent · Select child · Next sibling · Window / crossing · Rectangle region · Lasso region · Paint region

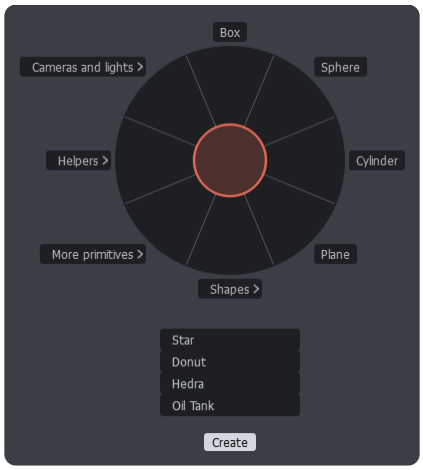
Changes with the situation (first match wins):

- **Selection** - when Edit Poly is current in the Modify panel, sub-object level: object level
- **Edit Poly: selection** - when Edit Poly is current in the Modify panel
- **Selection** - when the object is an Editable Poly, sub-object level: object level
- **Poly: selection** - when the object is an Editable Poly

## Alt+4 Create

dial 4

| | | |
|---|---|---|
| **NW**<br>Cameras and lights > | **N**<br>Box | **NE**<br>Sphere |
| **W**<br>Helpers > | **Create** | **E**<br>Cylinder |
| **SW**<br>More primitives > | **S**<br>Shapes > | **SE**<br>Plane |




**Submenu Shapes (S):** N Line · NE Arc · E Circle · SE Ellipse · S Rectangle · SW NGon · W Helix · NW Text

**Submenu More primitives (SW):** N Cone · NE Torus · E Tube · SE GeoSphere · S Pyramid · SW Teapot · W Chamfer Box · NW Capsule

**Submenu Helpers (W):** N Dummy · NE Point · E Grid · SE Tape · S Protractor · SW Compass · W Expose Transform · NW Container

**Submenu Cameras and lights (NW):** N Target camera · NE Physical camera · E Free camera · SE Sun positioner · S Free light · SW Target light · W Omni light · NW Match camera to view

**List under the dial:** Star · Donut · Hedra · Oil Tank

Changes with the situation (first match wins):

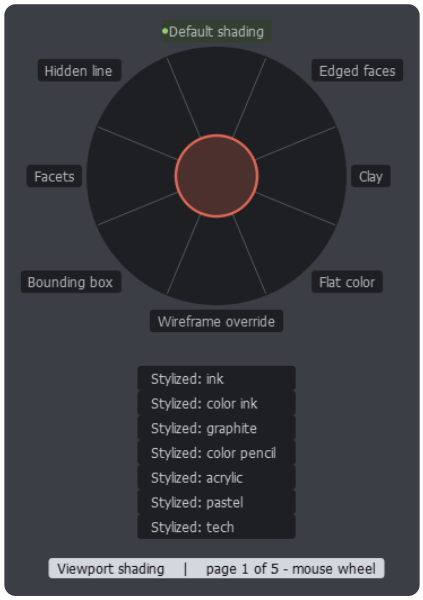
- **Create shapes** - when a shape (spline) is selected

## Alt+5 Viewport shading

dial 5

| | | |
|---|---|---|
| **NW**<br>Hidden line | **N**<br>Default shading | **NE**<br>Edged faces |
| **W**<br>Facets | **Viewport shading** | **E**<br>Clay |
| **SW**<br>Bounding box | **S**<br>Wireframe override | **SE**<br>Flat color |

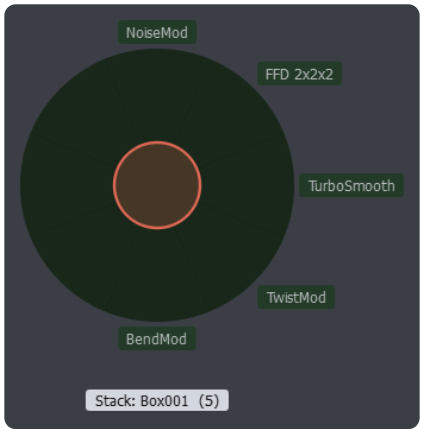
**List under the dial:** Stylized: ink · Stylized: color ink · Stylized: graphite · Stylized: color pencil · Stylized: acrylic · Stylized: pastel · Stylized: tech




## Alt+6 Modifier stack

dial 6

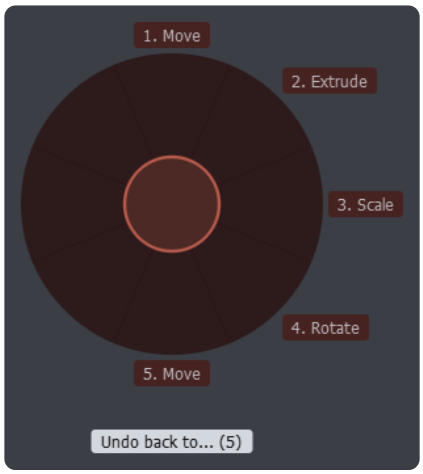
Filled live from the modifier stack of the selected object (top eight).




## Alt+7 Undo

dial 7

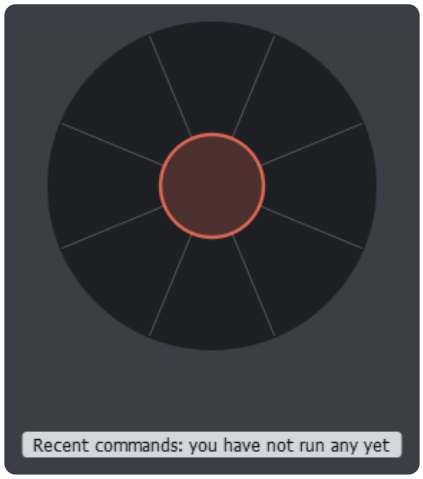
Filled live from Max's undo list - N undoes one step, NE two, and so on.



## Alt+8 Recent commands

dial 8

Filled live from the eight commands you ran most recently.

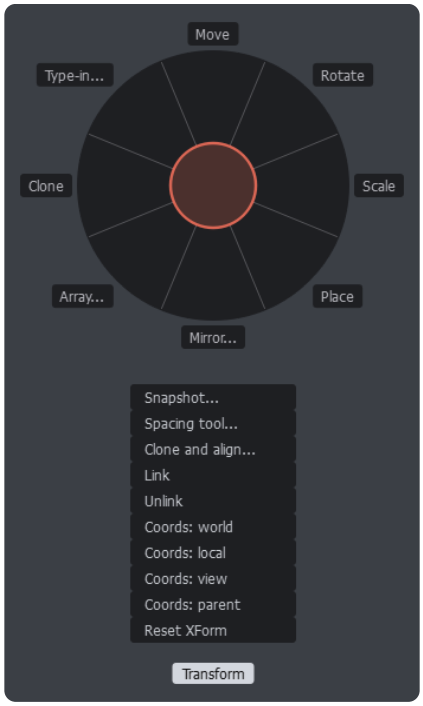



## Ctrl+Alt+1 Transform

dial 9

| | | |
|---|---|---|
| NW<br>Type-in... | N<br>Move | NE<br>Rotate |
| W<br>Clone | **Transform** | E<br>Scale |
| SW<br>Array... | S<br>Mirror... | SE<br>Place |

**List under the dial:** Snapshot... · Spacing tool... · Clone and align... · Link · Unlink · Coords: world · Coords: local · Coords: view · Coords: parent · Reset XForm

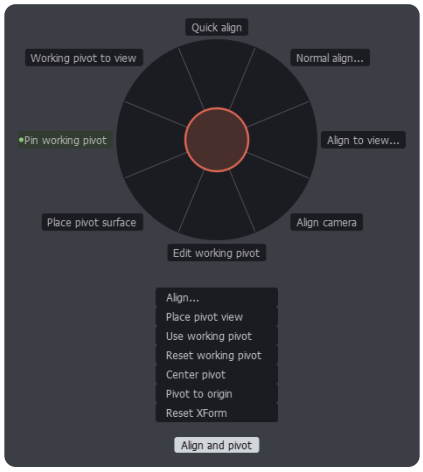



## Ctrl+Alt+2 Align and pivot

dial 10

| | | |
|---|---|---|
| NW<br>Working pivot to view | N<br>Quick align | NE<br>Normal align... |
| W<br>Pin working pivot | **Align and pivot** | E<br>Align to view... |
| SW<br>Place pivot surface | S<br>Edit working pivot | SE<br>Align camera |

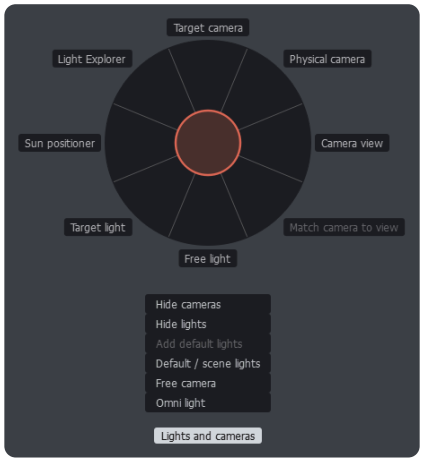
**List under the dial:** Align... · Place pivot view · Use working pivot · Reset working pivot · Center pivot · Pivot to origin · Reset XForm



## Ctrl+Alt+3 Lights and cameras

dial 11

| | | |
|---|---|---|
| **NW**<br>Light Explorer | **N**<br>Target camera | **NE**<br>Physical camera |
| **W**<br>Sun positioner | **Lights and cameras** | **E**<br>Camera view |
| **SW**<br>Target light | **S**<br>Free light | **SE**<br>Match camera to view |




**List under the dial:** Hide cameras · Hide lights · Add default lights · Default / scene lights · Free camera · Omni light

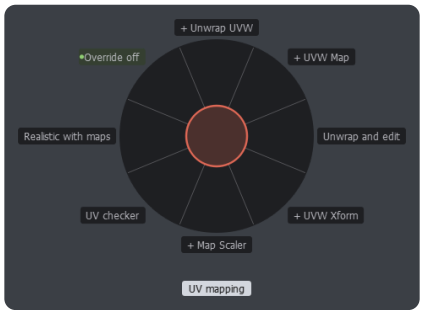
Changes with the situation (first match wins):

- **Camera** - when a camera is selected
- **Light** - when a light is selected

## Ctrl+Alt+4 UV mapping

dial 12

| | | |
|---|---|---|
| **NW**<br>Override off | **N**<br>+ Unwrap UVW | **NE**<br>+ UVW Map |
| **W**<br>Realistic with maps | **UV mapping** | **E**<br>Unwrap and edit |
| **SW**<br>UV checker | **S**<br>+ Map Scaler | **SE**<br>+ UVW Xform |




Changes with the situation (first match wins):

- **UV (Unwrap)** - when Unwrap UVW is current in the Modify panel

## Ctrl+Alt+5 Snaps and grid

dial 13

| | | |
|---|---|---|
| **NW**<br>Grid front | **N**<br>Snaps | **NE**<br>Angle snap |
| **W**<br>Grid top | **Snaps and grid** | **E**<br>Percent snap |
| **SW**<br>AutoGrid | **S**<br>Hide grids | **SE**<br>Snap settings... |

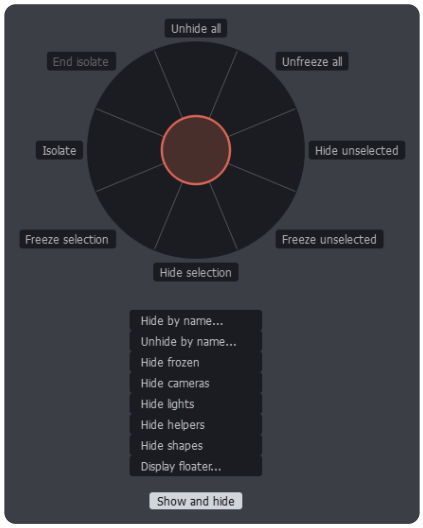



**List under the dial:** Cycle snap type · Snap to frozen · Axis constraints in snaps · Spinner snap · Grid left · Home grid

## Ctrl+Alt+6 Show and hide

dial 14

| | | |
|---|---|---|
| **NW**<br>End isolate | **N**<br>Unhide all | **NE**<br>Unfreeze all |
| **W**<br>Isolate | **Show and hide** | **E**<br>Hide unselected |
| **SW**<br>Freeze selection | **S**<br>Hide selection | **SE**<br>Freeze unselected |




**List under the dial:** Hide by name... · Unhide by name... · Hide frozen · Hide cameras · Hide lights · Hide helpers · Hide shapes · Display floater...

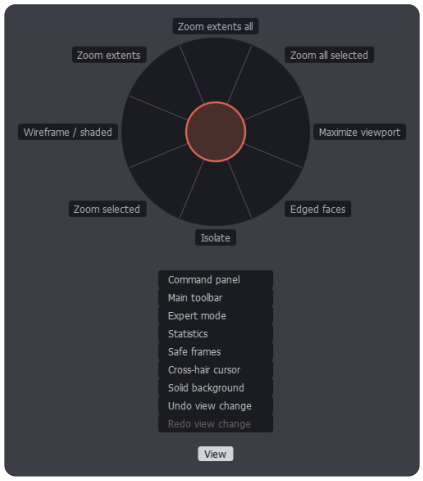
Changes with the situation (first match wins):

- **Show and hide** - when the object is an Editable Poly, sub-object level: object level
- **Poly: show and hide** - when the object is an Editable Poly

## Ctrl+Alt+7 View

dial 15

| | | |
|---|---|---|
| **NW**<br>Zoom extents | **N**<br>Zoom extents all | **NE**<br>Zoom all selected |
| **W**<br>Wireframe / shaded | **View** | **E**<br>Maximize viewport |
| **SW**<br>Zoom selected | **S**<br>Isolate | **SE**<br>Edged faces |




**List under the dial:** Command panel · Main toolbar · Expert mode · Statistics · Safe frames · Cross-hair cursor · Solid background · Undo view change · Redo view change

## Ctrl+Alt+8 Viewports

dial 16

| | | |
|---|---|---|
| **NW**<br>Camera | **N**<br>Top | **NE**<br>Perspective |
| **W**<br>Left | **Viewports** | **E**<br>Right |
| **SW**<br>Back | **S**<br>Bottom | **SE**<br>Front |

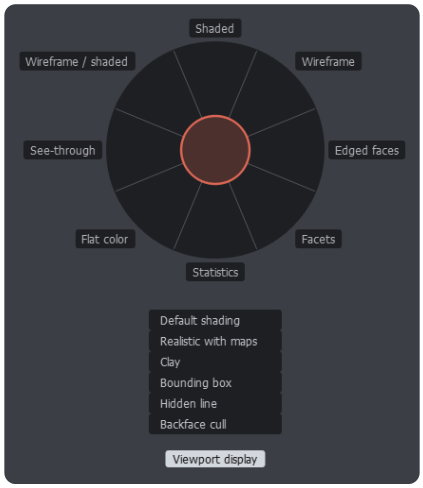



**List under the dial:** Orthographic · Maximize viewport · Home view · Save view · Restore view · Match camera to view · Viewport configuration...

## Shift+Alt+1 Viewport display

dial 17

| | | |
|---|---|---|
| **NW**<br>Wireframe / shaded | **N**<br>Shaded | **NE**<br>Wireframe |
| **W**<br>See-through | Viewport display | **E**<br>Edged faces |
| **SW**<br>Flat color | **S**<br>Statistics | **SE**<br>Facets |

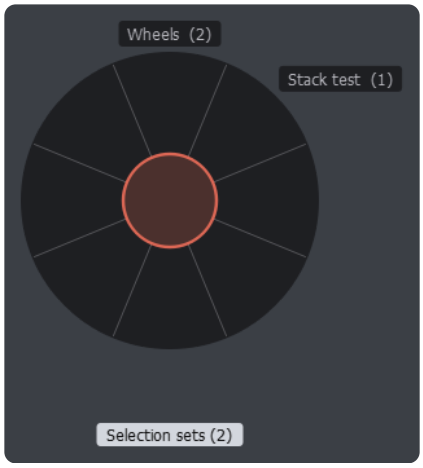



**List under the dial:** Default shading · Realistic with maps · Clay · Bounding box · Hidden line · Backface cull

## Shift+Alt+2 Selection sets

dial 18

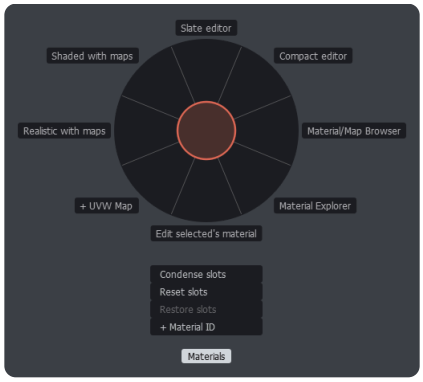
Filled live from the named selection sets of the scene.




## Shift+Alt+3 Materials

dial 19

| | | |
|---|---|---|
| **NW**<br>Shaded with maps | **N**<br>Slate editor | **NE**<br>Compact editor |
| **W**<br>Realistic with maps | Materials | **E**<br>Material/Map Browser |
| **SW**<br>+ UVW Map | **S**<br>Edit selected's material | **SE**<br>Material Explorer |




**List under the dial:** Condense slots · Reset slots · Restore slots · + Material ID

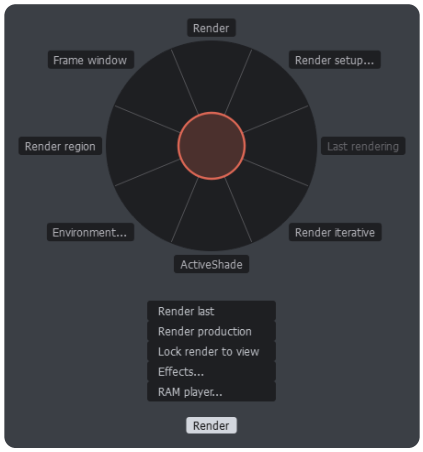
Changes with the situation (first match wins):

- **Materials (Slate)** - when cursor over the Slate Material Editor

## Shift+Alt+4 Render

dial 20

| | | |
|---|---|---|
| **NW**<br>Frame window | **N**<br>Render | **NE**<br>Render setup... |
| **W**<br>Render region | **Render** | **E**<br>Last rendering |
| **SW**<br>Environment... | **S**<br>ActiveShade | **SE**<br>Render iterative |

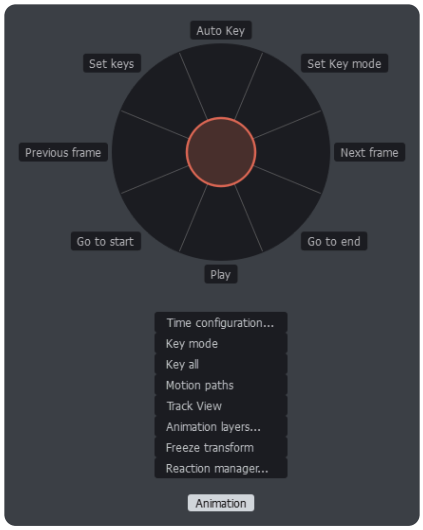



**List under the dial:** Render last · Render production · Lock render to view · Effects... · RAM player...

## Shift+Alt+5 Animation

dial 21

| | | |
|---|---|---|
| **NW**<br>Set keys | **N**<br>Auto Key | **NE**<br>Set Key mode |
| **W**<br>Previous frame | **Animation** | **E**<br>Next frame |
| **SW**<br>Go to start | **S**<br>Play | **SE**<br>Go to end |




**List under the dial:** Time configuration... · Key mode · Key all · Motion paths · Track View · Animation layers... · Freeze transform · Reaction manager...

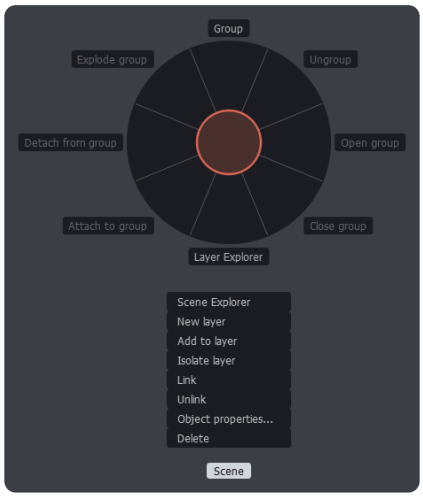
Changes with the situation (first match wins):

- **Track View** - when cursor over Track View

## Shift+Alt+6 Scene

dial 22

| | | |
|---|---|---|
| **NW**<br>Explode group | **N**<br>Group | **NE**<br>Ungroup |
| **W**<br>Detach from group | **Scene** | **E**<br>Open group |
| **SW**<br>Attach to group | **S**<br>Layer Explorer | **SE**<br>Close group |




**List under the dial:** Scene Explorer · New layer · Add to layer · Isolate layer · Link · Unlink · Object properties... · Delete

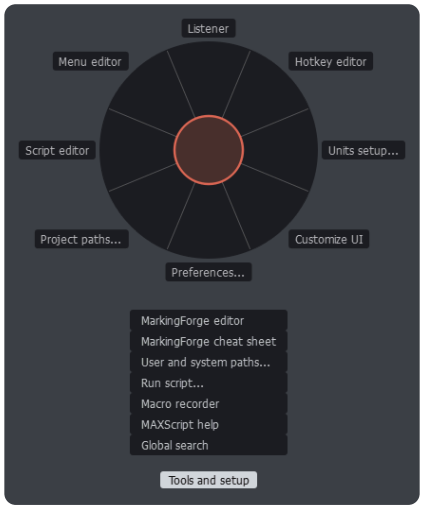
Changes with the situation (first match wins):

- **Scene Explorer** - when cursor over the Scene Explorer

## Shift+Alt+7 Tools and setup

dial 23

| | | |
|---|---|---|
| **NW**<br>Menu editor | **N**<br>Listener | **NE**<br>Hotkey editor |
| **W**<br>Script editor | **Tools and setup** | **E**<br>Units setup... |
| **SW**<br>Project paths... | **S**<br>Preferences... | **SE**<br>Customize UI |




**List under the dial:** MarkingForge editor · MarkingForge cheat sheet · User and system paths... · Run script... · Macro recorder · MAXScript help · Global search

## Shift+Alt+8 File

dial 24

| | | |
|---|---|---|
| **NW**<br>Hold | **N**<br>Save | **NE**<br>Save as... |
| **W**<br>Merge... | **File** | **E**<br>Save incremental |
| **SW**<br>Import... | **S**<br>Open... | **SE**<br>Save copy as... |




**List under the dial:** Fetch · Export... · Export selected... · Save selected... · Archive... · Summary info... · Reset

# 12. Files and settings

The program and your settings live apart: installing, updating and uninstalling change only the plugin folder, never the settings folder.

| What | Where |
|---|---|
| The plugin | `%APPDATA%\Autodesk\ApplicationPlugins\MarkingForge` - or `%ProgramData%\Autodesk\ApplicationPlugins\MarkingForge` when installed for all users |
| The dial library, the guides | in the plugin folder, `dials\` and `Documentation\` (**MarkingForge > Documentation** opens the guides) |
| The preset packs | in the plugin folder, `presets\MarkingForge_Pack_*.json` (Modelling, Animation, Look-dev, Archviz) - the editor's **Load preset...** opens it; the same files are in `Presets\` of the download |
| Your own presets | wherever you save them - **Save preset...** offers the settings folder below, `MarkingForge_preset.json` |
| Your settings folder | `%LOCALAPPDATA%\Autodesk\3dsMax\2027 - 64bit\ENU\en-US\plugcfg\MarkingForge` - for an English 3ds Max 2027; every 3ds Max version (and language) has a folder of its own |
| Your dials | that folder, `menus.json` - every dial, its sets, variants and colours |
| Your shortcuts | that folder, `hotkeys\MarkingForge_Hotkeys.hsx` - 3ds Max's active hotkey set, with ALL your shortcuts; `hotkeys\backup_*.hsx` - one before every change |
| Usage counters | that folder, `usage.json` |
| The editor's own colours | that folder, `editor_colors.json` - only once you choose some |
| Your colour presets | that folder, `dial_color_presets.json` (dials) and `editor_color_presets.json` (the editor) |
| Start-up log | that folder, `menu_log.txt` |
| First-start records | that folder: `setup_shortcuts.flag` or `skip_shortcuts.flag` (the installer's tick box, until the next start reads it) and `shortcuts_first_start.done` (the first start has set up the keys) |

**MarkingForge > Open configuration folder** opens the settings folder in Explorer - the sure way to find it whatever the language of your 3ds Max; **Open menus.json** opens the file in your text editor. The plugin notices a saved file and reloads it by itself.

## Keeping the dials in another folder

Set the Windows environment variable `MARKINGFORGE_CONFIG_DIR` to the FULL path of a folder that exists, then start 3ds Max: the plugin and the editor read and write menus.json and the files beside it - usage, shortcut files, colours, presets - in that folder instead. It is meant for a test copy that must never touch your own dials, or for a prepared studio set. A folder without a menus.json starts with the built-in dials.

```
set MARKINGFORGE_CONFIG_DIR=D:\MarkingForge_test
"%ProgramFiles%\Autodesk\3ds Max 2027\3dsmax.exe"      (or wherever your 3ds Max is installed)
```

In a command prompt or a .bat file as above it applies to that 3ds Max only; set in Windows' environment variables it applies to every 3ds Max started afterwards.

A relative path, or a folder that does not exist, is ignored and the usual folder is used; quotes around the path are taken off. **MarkingForge > Diagnostics > Configuration status** prints a *config dir* line saying which folder is in use and why. The start-up log and the first-start records stay in the usual folder, and the studio script does not follow the variable (it has `-SettingsRoot`).

To move your setup to another computer or another 3ds Max version: **Save every dial...** there and **Load every dial...** here (the dials, their sets and colours), or copy `menus.json` while 3ds Max is closed. **Save preset...** carries the dials, their pages and colours and leaves out paths and usage data; one colour preset travels with **Presets > Export this preset to a file...** Keyboard shortcuts are not in these files - on the other computer the first start or **Set up standard shortcuts...** gives them.

# 13. Updating, moving and uninstalling

## Updating

Install the new version the same way as the first time - any of the four methods. The Setup window shows **Update / Repair**; your dials, colours and shortcuts are kept, because they live in the settings folder, which no installer touches. Restart 3ds Max. With method 3, close 3ds Max and replace the whole `MarkingForge` folder rather than copying over it; with method 4, run the script again.

An update cannot leave you without a working plugin: the installer first backs up the installed version (into 3ds Max's temp folder; the studio script into Windows' TEMP), and if copying the new one fails - a full disk, an antivirus lock - it puts the previous version back and says so. Nothing to do but fix the cause and run it again. The installer also works while the old version is loaded: the loaded `MarkingForge.gup` is renamed to `MarkingForge.gup.old_<n>`, which 3ds Max does not load, and the next run removes it.

> **New built-in dials after an update.** Your `menus.json` is kept exactly as it was, so a dial that a new version changes does not replace yours - *CHANGELOG.txt* says what is new. **Load default dials...** brings the new ones to the dials that have no menu of their own; **Dial library...** > **Restore Menu N** puts one original back.

## Moving to another version of 3ds Max

Every 3ds Max version needs its own MarkingForge download and keeps its own settings folder (the path holds the version - `2027 - 64bit`). Install the download for the new version, then bring your dials over: **Save every dial...** in the old 3ds Max, **Load every dial...** and **Save** in the new one - or copy `menus.json` between the two settings folders while 3ds Max is closed. The first start of the new version gives the dials their standard keys.

## Uninstalling

1. Remove the plugin, in one of three ways:
   - run the installer (method 1 or 2), press **Uninstall** and answer **Yes** - it removes MarkingForge for the current user and for all users (the latter needs 3ds Max started as administrator); a file in use is renamed so 3ds Max will not load it - run Uninstall once more after the restart if it says *Not everything could be removed*;
   - with the studio script, 3ds Max closed: the same command as in method 4 with `-Uninstall -Scope AllUsers` (or `-Scope CurrentUser`) - it removes that one scope;
   - by hand: close 3ds Max and delete the plugin folder from chapter 12.

   The installer and the studio script remove only that folder (and their two flag files in the settings folder). A junction or symbolic link inside it - or the folder itself being one, say to a studio share - is removed as a link; what it points to is never touched (1.5.15).
2. Restart 3ds Max - it loads plugins only while it starts, so it lets go of MarkingForge only then.
3. Optional: delete the settings folder `plugcfg\MarkingForge` as well - but NOT its `hotkeys` subfolder: 3ds Max uses the shortcut file in it as your active shortcut set, with ALL your shortcuts, not only MarkingForge's. Assign another set in **Customize > Hotkey Editor** first if you want that folder gone too.

# 14. Troubleshooting

**Known 3ds Max behaviour: the object behind lights up while you edit sub-objects**
Seen in the 2027 version of 3ds Max, possible in other versions: while you edit vertices, edges or polygons of one object, the object **behind** it can light up under the cursor - when you press a dial's key or pan with the middle mouse button, also on an object without modifiers. It shows even during the middle-button pan, which MarkingForge does not touch. A dial works on the object you edit and does not select the one behind (since 1.5.79). If it gets in your way: tick **MarkingForge > Experimental features > No object outline in sub-object mode**, or switch off 3ds Max's *Preview Highlight Outlines Toggle* (Customize > Hotkey Editor, category Views). If the object behind did become selected, Ctrl+Z gives your selection back.

| Symptom | Cause and fix |
|---|---|
| **Install** is greyed out in the Setup window | A check in the status box failed. *THIS BUILD NEEDS 3ds Max 2027*: this download is for another 3ds Max version - get the one for yours. *FILES MISSING next to the installer*: the download was not fully unzipped, or `MarkingForge_Install.ms` was moved away from its `MarkingForge` folder. |
| The Setup window says *These files of the installed MarkingForge are open in another program* | A document of the installed version is open - a PDF in Acrobat, say. Close it and press Install (or Uninstall) again; nothing was changed. |
| The Setup window says *Copying FAILED* | For all users, 3ds Max has to be started as administrator. Otherwise a full disk or an antivirus lock - the previous version is put back; fix the cause and install again. |
| No MarkingForge menu after the restart | Type `MarkingForge.version()` in the Listener. "Unknown property" = not loaded. Check that the folder is in `ApplicationPlugins` (chapter 12) and contains `PackageContents.xml`; reinstall the Visual C++ 2015-2022 x64 runtime. If the plugin is loaded but the menu is missing, `menu_log.txt` in the settings folder says why. |
| 3ds Max reports a duplicate plugin at start | Another copy of `MarkingForge.gup` exists, usually in the 3ds Max `plugins` folder - or MarkingForge is installed both for you and for all users. The Setup window lists every copy it finds - delete all but one copy in ApplicationPlugins. |
| The menu editor's title says *OUT OF DATE* | 3ds Max is running an older plugin than the editor beside it: 3ds Max was not restarted after an update, or it loads another, older copy (see the row above). |
| The dials have no keys after installing | The first start sets them up only when no dial has a key yet, once per 3ds Max version - and never when the installer's tick box was cleared or the studio script ran with `-NoShortcuts`. Choose **MarkingForge > Set up standard shortcuts...** |
| A key does not open a dial | Check it in **Customize > Hotkey Editor**: another command may use the same key. Run **Set up standard shortcuts...** or set the key in the menu editor. |
| The mouse wheel zooms the viewport instead of turning pages | The dial has only one set - pages are its sets, made in the editor. The wheel also goes to 3ds Max in a submenu, in the hotbox and during a search. |
| The dials are not the ones you saved, or are the built-in ones again | Check **Diagnostics > Configuration status**: a *config dir* line means `MARKINGFORGE_CONFIG_DIR` points 3ds Max at another folder (chapter 12) - or says why it was ignored. |
| One shortcut never reaches 3ds Max (e.g. Ctrl+Alt+5), or | Another program has taken that key combination for the whole of Windows, and Windows gives it to that program before 3ds Max sees it - no setting inside 3ds Max can change that. Common owners: **PowerToys** (ZoomIt uses Ctrl+Alt+digits), graphics-driver overlays (NVIDIA, AMD, Intel), Discord, screenshot tools, mouse and keyboard software (Logitech, Razer). Find the program by pressing the key while 3ds Max is in front and |

| | |
|---|---|
| pressing it brings up another program | watching which window appears, then change or switch off the shortcut in that program - or give the dial another key in the menu editor. Since 1.5.22 the menu editor says so in red under the dial's shortcut, the **Set shortcut...** window warns before you take such a key, and **Fill the free slots...** (and the first start) leaves it out. It sees programs that register the key with Windows (ZoomIt, overlays, screen recorders), not tools that read the keyboard by other means (AutoHotkey). |
| No shortcut works - no dial, and none of 3ds Max's own either - until you click a viewport | 3ds Max itself has switched its shortcuts off: measured after starting a **Text** object (TextPlus), whose typing takes the keyboard. Click in a viewport or press Escape. A dial whose release 3ds Max fails to report in such a state closes by itself a quarter of a second after you let go (1.4.4); `MarkingForge.eventLog()` then shows `RELEASE_LOST`. |
| A dial opens empty | It uses a live source with nothing to show (no modifiers, no selection sets). The cheat sheet lists such dials at the top. |
| Save says the configuration changed after it was opened | menus.json was changed on disk while the editor was open (another editor window, or a tick in MarkingForge > Experimental features). Save your edits as a separate copy, press **Reload from file** and carry the edits over. |
| The dial shows a different menu than expected | A contextual variant matched. `MarkingForge.eventLog()` names the variant that won. |
| In a queue, the command after Inset (Extrude, Bevel...) does not start | Finish the tool with a **right-click** (or Q, W, E, R) - Escape keeps it on in 3ds Max. Picking another tool, selecting another object, a new gesture or a new scene drops the rest of the queue on purpose; `MarkingForge.eventLog()` shows a `QUEUE` line saying why. |
| Holding a dial key, you switched to another program; the next press of that key does nothing | 3ds Max did not see the key being released in the other program (measured in 2025, 2026 and 2027) and ignores the next press. Press the digit once on its own (for Alt+1: just **1**) - or simply press the dial key again; the press after works. |
| The editor does not open | The error is shown in a message; also check that the `tools` folder is inside the plugin folder. |
| A menus.json edited by hand stopped working | **MarkingForge > Diagnostics > Configuration status** names the line with the error. A broken item disables only that item. |

## Listener commands

```
MarkingForge.version()          -- is the plugin loaded
MarkingForge.configPath()       -- which menus.json is in use
MarkingForge.configSummary()    -- errors and warnings in menus.json
MarkingForge.eventLog()         -- the last gestures: what opened, what ran
MarkingForge.windowInfo()       -- what the plugin sees under the cursor
MarkingForge.mouseInfo()        -- mouse-button bindings
```

MarkingForge sends nothing over the network, does not touch the keyboard outside 3ds Max and changes your shortcuts only when you ask it to.

> **Still stuck?** Open an issue at github.com/Forge-Plugins/MarkingForge/issues, ask on the Discord channel or write to forgeplugins@gmail.com. **MarkingForge > Report a bug...** puts your 3ds Max and MarkingForge details on the clipboard, ready to paste.

# 15. Support, bug reports and feature requests

| | |
|---|---|
| ISSUES | github.com/Forge-Plugins/MarkingForge/issues |
| ROADMAP | trello.com/b/7gJVqG8Y/markingforge-features-roadmap |
| DISCORD | discord.gg/HsakZWQZAw (questions, tips, news - with other users and the author) |
| E-MAIL | forgeplugins@gmail.com |
| WEB | https://forgeplugins.tech (forgeplugins.tech - more plugins by the author) |

**Bug reports and feature requests** are best opened as issues on GitHub - the forms there ask for exactly what is needed, and you can follow the fix. Quick questions and tips: the **Discord** channel (**MarkingForge > Discord community** opens it). Anything you would rather not post publicly (licence, purchase) goes to the e-mail address. Every message is read.

## Reporting a bug

1. In 3ds Max choose **MarkingForge > Report a bug...** - it copies your 3ds Max version, MarkingForge version, configuration summary and start-up log to the clipboard. It works even when the plugin did not load.
2. Open github.com/Forge-Plugins/MarkingForge/issues and pick **Bug report** - or write an e-mail to forgeplugins@gmail.com instead. Both are links in the window that opens (since 1.5.20), one click each.
3. Say what you did, what you expected and what happened instead, then paste the details with **Ctrl+V**.
4. Add a screenshot if you can. When the problem is in one dial, attach a preset of it (editor > **Save preset...**).

## Suggesting a feature

Open an issue with the **Feature request** form, choose **MarkingForge > Suggest a feature...**, or simply write. Describe what you would like to do and how you do it today - a concrete example from your work says more than a list of wishes. Ideas for new built-in dials, commands and options are all welcome. What is already planned and in progress is on the **Roadmap** above - **MarkingForge > Roadmap (Trello)** opens it too.

Nothing is sent by the plugin itself: the details go to the clipboard and your own e-mail program, so you see everything that leaves your computer. MarkingForge has no network access.